

Deep Learning-Enhanced Quantum State Estimation of Simulated Multi-Qubit Systems

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Recovering complete information about a quantum state is a fundamental challenge in quantum information science, owing to the probabilistic nature of quantum measurement, the impossibility of determining a state from a single measurement, and noise from imperfect state preparation and detection in an experiment. In this work, a multilayer perceptron neural network (NN) is developed for quantum state estimation, using supervised deep learning to map synthetic projective measurement data to the density matrices that fully characterise the corresponding quantum states. A suitable NN architecture is first identified through hyperparameter tuning, including Bayesian optimisation of the learning rate and regularisation strength. The feasibility of deep learning for quantum state tomography is then established in an abstract two-qubit setting by benchmarking the NN against two traditional approaches: Stokes reconstruction and maximum-likelihood estimation (MLE). Their relative noise-robustness is tested under independent phenomenological noise channels and, subsequently, scalability is assessed up to four qubits under a combined trapped ion-inspired noise condition derived from the reported error budget of a $^{40}\text{Ca}^+$ processor. Fidelity and inference time are investigated as key performance metrics; in the final four-qubit scalability test, the NN achieves a mean fidelity of $\bar{F} = 0.950$, corresponding to improvements of 20% over Stokes reconstruction and 33% over MLE. Furthermore, it reconstructs the full 1000-state test set in 0.26 s - a speedup by a factor of approximately 5 times over Stokes reconstruction and more than 8×10^3 times over MLE, demonstrating significant performance gains in both accuracy and computational cost. These results identify deep learning as a scalable solution to the limitations of traditional reconstruction methods, increasingly important in supporting the rapid growth of quantum technologies. More broadly, this research establishes an extensible methodological framework, laying a strong foundation for future adaptation and deployment to broad quantum state estimation and classification tasks.



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1. INTRODUCTION

A single measurement is insufficient to fully characterise a quantum system because it projects the wavefunction onto a single eigenstate of the observable being measured, irreversibly destroying the original superposition [1]. Quantum state tomography (QST) is the process of estimating the full quantum-mechanical description of a state, described by its density matrix ρ , using experimental data from measurements on many identically prepared copies of it [2]. Traditional QST methods, such as Maximum-Likelihood Estimation (MLE) and Stokes Reconstruction, are well established and reliable when sufficient data are available [3, 4]. However, their performance is limited by two major challenges. The exponential growth of the Hilbert space dimension, and hence the number of measurements required for full QST, with the number of qubits makes complete state reconstruction computationally demanding, and their sensitivity to noise induced by both the interaction of the system with its environment and the intrinsic uncertainty of quantum measurement that distorts the measured state [1]. Machine learning (ML) has consistently proved to be an efficient and effective alternative to human knowledge-based approaches [5], stimulating a growing interest in its use for quantum information tasks. This work explores the potential of ML-based QST as a scalable, noise-robust approach for accurate quantum state estimation.

Deep learning (DL) is a subset of ML in which layers of a neural network learn a mapping from input data to a target output. A major advantage of deep networks is their ability to learn intricate, high-dimensional, non-linear patterns and features from such data which can generalise to accurate predictions on similar, unseen data. This capability has been demonstrated across a wide range of tasks, including regression and classification, and DL has dramatically improved the state-of-the-art in speech recognition and other domains such as drug discovery [6]. In the realm of quantum state estimation, DL has proved to be effective at extracting information from high-dimensional and noisy probabilistic measurement data to accurately characterise the density matrix [4, 7, 8, 9, 10].

In this work I develop and train a deep neural network for quantum state estimation. I use synthetic measurement data under simulated noisy experimental conditions to train the model, and benchmark its performance against traditional physics-based methods. The key metrics investigated are fidelity, which quantifies the accuracy of the reconstructed state compared to the ideal target state, and inference time, which determines how efficiently the methods can infer information about states of exponentially increasing system sizes. The project is primarily a methodological study, and the proposed framework is designed to generalise to a range of real experimental settings and physical systems.

Three primary research questions are addressed in this work: (i) whether a deep neural network can feasibly perform quantum state estimation in an abstract setting; (ii) whether this approach can be applied to realistic physical systems by remaining robust to noisy and finite measurement data; and (iii) how its performance scales with increasing system size. The final question is explored under an error budget inspired by a trapped-ion quantum processor, serving as a well-motivated and state-of-the-art example of the conditions under which scalable QST is performed in practice [11]. The study ultimately seeks to determine whether DL-based QST can consistently outperform and address the limitations of traditional methods, thereby motivating its extension to real-world experimental settings.

2. THEORETICAL FRAMEWORK & BACKGROUND

A. Quantum State Preparation & Measurement

Qubits are two-level quantum systems that can exist in superpositions of the computational basis states $|0\rangle$ and $|1\rangle$. The state of an n -qubit system resides in a 2^n -dimensional Hilbert space, $\mathcal{H}_{2^n}$. Complete information about a quantum state is contained in its density matrix, $\rho \in \mathbb{C}^{d \times d}$. For a complex state vector $|\psi\rangle \in \mathbb{C}^d$, a pure state is represented by:

$$\rho = |\psi\rangle \langle \psi|. \quad (1)$$

A physically valid density matrix must satisfy three conditions [12]: (i) Hermiticity, $\rho^\dagger = \rho$; (ii) positive semi-definiteness (PSD), $\rho \succeq 0$; and (iii) unit trace, $\text{Tr}(\rho) = 1$. Ideal QST aims to reconstruct ρ subject to these constraints from an informationally complete set of measurements.

Quantum measurements are described by a set $\{E_m\}$ of measurement operators forming a positive operator-valued measure (POVM). Each POVM element E_m is a PSD matrix that satisfies the completeness relation (i.e. $E_m \geq 0$ and $\sum_m E_m = I$) and is associated with a potential measurement outcome m [1, 13]. The probability p_m of detecting outcome m from measurement of the quantum state ρ is given by the Born rule [14]:

$$p_m = \text{Tr}(\rho E_m), \quad (2)$$

with normalisation $\sum_m p_m = 1$ following from $\text{Tr}(\rho) = 1$.

For measuring qubits, the POVM is derived from the Pauli matrices $\{\sigma_1, \sigma_2, \sigma_3\}$ [15]. Together with the identity σ_0 , these are defined as:

$$\sigma_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (3)$$

These operators form an orthonormal basis for the space of single-qubit operators, and hence provide an informationally complete measurement basis for quantum state estimation. Extending to an n -qubit system, the Pauli basis is obtained from n -fold tensor products of the Pauli matrices [16]:

$$P_{\mathbf{k}} = \sigma_{k_1} \otimes \sigma_{k_2} \otimes \cdots \otimes \sigma_{k_n}, \quad k_j \in \{0, 1, 2, 3\}, \quad (4)$$

where $\mathbf{k}$ indexes the $d^2 = 4^n$ Pauli operators that span the space of density matrices.

Each of the $4^n - 1$ non-identity Pauli operators has eigenvalues ± 1 , and its measurement corresponds to a two-outcome POVM $\{E_{\mathbf{k}}^\pm\}$ with elements $E_{\mathbf{k}}^\pm = \frac{1}{2}(I \pm P_{\mathbf{k}})$ [12]. There are therefore $2(4^n - 1)$ POVM elements in total - the two-qubit set is presented graphically in Appendix A. A single measurement of a system yields either outcome $+1$ or -1 , corresponding to projection onto the respective eigenstates, with probabilities $p_{\mathbf{k},\pm}$ determined by Eq.(2). Repeated measurements, called shots, allow estimation of the expectation values $\langle P_{\mathbf{k}} \rangle$ which are the Pauli observables that fully characterise the density matrix.

In practice, ideal QST is limited by quantum state preparation and measurement (SPAM) errors [17]. First, the probabilistic nature of quantum measurements requires infinite shots for perfect state reconstruction, so projection noise is introduced by taking a finite number of shots.

In addition, imperfections in state preparation, control, and detection, such as thermal fluctuations or readout failures, can perturb the measurement outcomes from their true expectation values or cause measurements to be uninformative, leading to a non-ideal state reconstruction.

These limitations become increasingly significant as system size grows. Recalling $d = 2^n$, the number of measurements required to characterise the quantum state scales exponentially with the number of qubits. Consequently, SPAM errors and computational cost are amplified in larger systems.

B. Motivation and Physical Applications

In quantum information science - encompassing quantum computation [1], communication [18], metrology [19], and quantum optics [20] - accurately characterising quantum states is pivotal in validating state preparation, diagnosing noise, and benchmarking quantum devices [10]. QST has become the standard tool for achieving this task across broad fields.

The earliest proposed technique was optical homodyne tomography to determine the density matrix of a single light mode from repeated quadrature measurements at varying phases, successfully demonstrated for squeezed vacuum states [21]. The success of homodyne tomography stimulated the development of QST procedures for a range of physical systems, including photonic systems in which qubits are encoded in polarisation states [3, 17], and ions confined in a Paul trap [11, 22], where qubits are encoded in the internal energy levels of the ion. Trapped-ion processors have emerged as a particularly promising realisation of a quantum computer (QC) [23], where the verification of hardware relies on accurate quantum state estimation.

QCs replace binary classical bits with qubits, exploiting their quantum mechanical properties to solve certain classes of problems intractable for classical computers [1]. The key phenomena that underlie this so-called quantum advantage are:

a. Superposition & Interference A qubit simultaneously holds amplitudes for $|0\rangle$ and $|1\rangle$, expressed as:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle \quad (5)$$

for probability amplitudes $\alpha, \beta \in \mathbb{C}$. This enables n qubits to explore exponentially large state spaces in $\mathcal{H}_{2^n}$ in parallel.

b. Entanglement A multi-qubit state is entangled if it cannot be factorised into a tensor product of individual qubit states. In a QC, entanglement is exploited to create non-classical correlations, even when the individual states are far apart, that underpin quantum gate operations and the speedup of quantum algorithms [24]. Each measurement in an entangled multi-qubit system collapses the global wavefunction, reflecting the intricate correlations that entanglement provides. For examples of maximally-entangled two-qubit Bell states [25] and highly-entangled n -qubit W states, see Appendix C.

Quantum computations are implemented as unitary circuits U_ψ acting on an initial state $|\psi\rangle$. These circuits are composed of quantum gates, including single-qubit rotations (e.g. $R_X(\theta)$, $R_Y(\theta)$, $R_Z(\theta)$) together with two-qubit entangling gates such as the $R_{XX}(\theta)$ or CNOT. Rotations allow measurements in alternative bases and hence provide the informational completeness required for full QST, while entangling gates generate the non-classical correlations central

to multi-qubit state preparation and quantum advantage. Together, these operations form a universal gate set of which any n -qubit unitary can be decomposed [26].

While the methods developed in this work are intended to generalise to broad physical systems, quantum computing provides a particularly well-motivated and illustrative application. It is especially relevant to the present study both as the basis for the state-generation procedure and the derivation of the noise condition for the realistic operating point used to benchmark scalability in response to research question (iii). As the number of qubits increases, processors access to exponentially large Hilbert spaces to solve more complex problems, but this also makes state full state characterisation exceedingly computationally demanding. This is compounded by the accumulation of SPAM errors in larger systems. Among the approaches proposed to address this [27], this research investigates whether deep learning can provide an efficient, noise-robust, and scalable quantum state estimation method.

C. Deep Neural Networks

Artificial intelligence (AI) is the development of algorithms capable of performing tasks that typically require human-level intelligence. Machine learning (ML) is a subset of AI in which a computer improves its performance on a task by learning from data rather than being explicitly programmed [28, 29]. This makes ML algorithms excellent candidates for real-world problems whose scale or complexity is beyond feasible human capability. Its applications span a wide range of domains, including fraud detection, financial forecasting [30], medical diagnosis [31], autonomous driving [32] and personalised recommendation engines [33].

Deep learning (DL) extends conventional ML by using neural networks with many layers of interconnected neurons, loosely inspired by the structure of the brain. At its core, a neural network is a parameterised function that learns a mapping from input data to a target output. This layered architecture, called a multi-layer perceptron (MLP), enables neural networks to capture high-dimensional and non-linear structure, making them especially powerful in prediction and classification tasks such as speech and handwritten digit recognition [34].

Neurons are the building blocks of deep learning architectures. The operation of a neuron is mathematically represented as:

$$y = f \left(\sum_{i=1}^n w_i x_i + b \right), \quad (6)$$

where n is the number of neurons in the previous layer. The key features are:

- x_i : input values, represented by neurons in the previous layer;
- w_i : weights, which determine the strength and sign of the influence of each input;
- b : bias term added to the weighted sum, which shifts the threshold for neuron activation;
- f : activation function that introduces non-linearity, allowing the network to learn complex patterns and relationships in the data [35].

During training, weights and biases are adjusted to minimise a loss function, which measures the discrepancy between the network prediction and target output values. This minimisation is carried out by an optimiser algorithm with a step size defined by its learning rate.

This research adopts a supervised learning framework, in which the network is trained using examples of both inputs and their known outputs (labels). In this context, the input layer

contains measurement data and the output layer represents the density matrix elements of the corresponding quantum state. Training is performed by stochastic gradient descent via the back-propagation algorithm, which systematically applies the chain rule to compute the gradient of the loss function with respect to each trainable parameter in the network. Mini-batch training is a method to reduce computational cost by performing this algorithm on representative subsets of the training set. Over repeated training epochs, the network predicts the output, evaluates the loss, and updates its weights and biases accordingly to converge to the target. Once trained, the network is tested on previously unseen data by predicting outputs using the parameters it has learned during training.

A key challenge in training a neural network is overfitting, where the model learns features specific to the training data, including noise, to an extent that it cannot faithfully generalise to unseen examples. Regularisation techniques mitigate this effect and are discussed in the methodology. The choice of activation function, loss function, and optimiser algorithm can also strongly influence model performance and convergence rate. A broader overview of neural network architectures and commonly used functions is provided in Appendix B.

More advanced DL architectures, such as convolutional neural networks (CNNs) [36], have been highly effective in processing grid-structured data such as images [37]. However, since the input data here consist of numerical measurement vectors, an MLP provides a suitable and flexible architecture for QST. Initial results were sufficiently strong that the focus was placed on optimising the MLP through hyperparameter tuning. While CNN-based approaches could be a direction for future work, existing results in QST [9] did not motivate their inclusion here.

3. METHODOLOGY

A. Dataset Generation

1. Unitary Circuit Model

The first stage of this investigation is to generate a sufficiently large and diverse ensemble of valid, known, pure quantum states from which QST performance can be evaluated statistically. State preparation is modelled using a randomised unitary quantum circuit, providing a physically motivated and scalable framework for the purposes of a synthetic experiment. The specific circuit model does not itself dictate the object of study, and the computational framework can be extended to other state classes, including Haar-random pure states and Ginibre mixed states [38], as outlined in Appendix C.

The n -qubit register is initialised in the computational ground state,

$$|\psi_0\rangle = |0\rangle^{\otimes n} = |00 \cdots 0\rangle. \quad (7)$$

A randomised unitary circuit is then applied, consisting of alternating layers of global single-qubit rotations and an all-to-all XX -type entangling operation. The total unitary acting on the

register is written as:

$$U_\psi = \prod_{\ell=1}^D \left[\underbrace{\bigotimes_{i=1}^n R_X(\theta_X^{(\ell)}) R_Y(\theta_Y^{(\ell)})}_{\text{global single-qubit rotations}} \underbrace{U_{XX}(\theta_{XX}^{(\ell)})}_{\text{entangling gate}} \right] \quad (8)$$

This ansatz is inspired by ion-trap gate structures [39], but is used here primarily as a flexible synthetic state generator. At each layer ℓ :

- $R_Y(\theta_Y)$: Rotates qubit i about the Y -axis by an angle $\theta_Y \sim \mathcal{U}(0, 2\pi)$.
- $R_X(\theta_X)$: Rotates qubit i about the X -axis by an angle $\theta_X \sim \mathcal{U}(0, 2\pi)$.
- $U_{XX}(\theta_{XX})$: Applies an effective entangling rotation with $\theta_{XX} \sim \mathcal{U}(0, \pi/2)$.

The circuit depth D is sampled from $D \sim \mathcal{U}\{1, 2, 3\}$. For each generated state, all gate angles and the circuit depth are resampled independently from their respective distributions. Since these parameters are continuous, the probability of generating two identical states is almost surely zero, ensuring a diverse ensemble under the gate set. Extended definitions of the individual gates and their computational implementation are given in Appendix C.

This operator evolves the initial register into a final pure state:

$$|\psi\rangle = U_\psi |\psi_0\rangle, \quad \rho = |\psi\rangle \langle\psi| \quad (9)$$

This guarantees that each generated state is physically valid and pure, since the initial state is normalised and all circuit operations are unitary. A dataset of N independent states is then generated by Monte Carlo sampling, with the circuit parameters redrawn for each example using a seeded pseudo-random number generator. The resulting ensemble spans a broad range of state structures making it suitable for robust statistical evaluation of the QST methods. An example of a generated two-qubit pure state is shown in Figure 1.

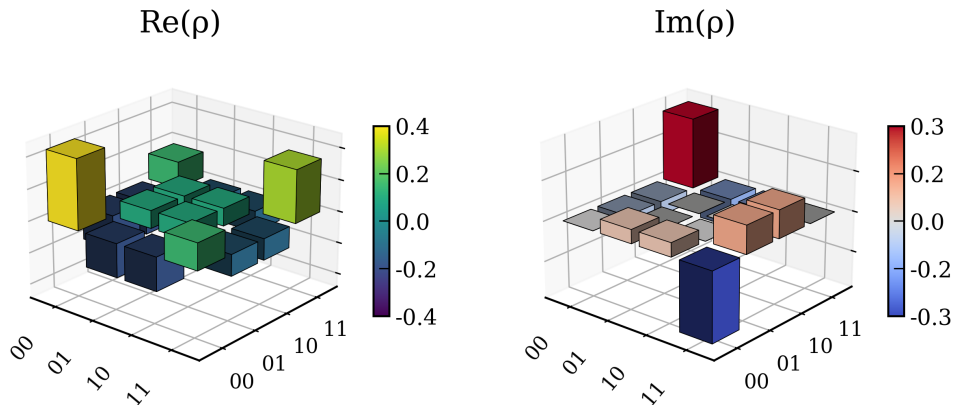


FIG. 1: Example target density matrix for a generated two-qubit pure state. Diagonal elements represent computational basis state populations, and are hence real and positive, and off-diagonal elements are coherences encoding relative phases between basis states.

2. Projective Measurements and Simulated Counts

For each ρ in the dataset, an informationally complete set of measurements is simulated by evaluating the expectation value of every n -qubit Pauli operator, as introduced in Section 2 A. The ideal Pauli expectations, or Stokes parameters, are computed directly from the density matrix as:

$$S_k = \langle P_k \rangle = \text{Tr}(\rho P_k), \quad k = 0, 1, \dots, 4^n - 1, \quad (10)$$

where $S_0 = \text{Tr}(\rho) = 1$ is always fixed by normalisation. Together these form the Stokes vector $\mathbf{S} \in \mathbb{R}^{4^n}$, which fully parameterises ρ through the expansion:

$$\rho = \frac{1}{d} \sum_{k=0}^{4^n-1} S_k P_k. \quad (11)$$

In experiment, however, these ideal expectation values are not directly accessible. Recall for each Pauli operator P_k , measurement corresponds to a projection onto either the $+1$ or -1 eigenspaces via the two-outcome POVM $\{E_k^\pm\}$ with a probability $p_{k,\pm}$ governed by the Born rule in Eq.(2). Each Pauli measurement is therefore simulated as a binomial sampling process over a finite number of shots N_{shots} , drawing counts of each outcome from [12]:

$$n_{k,+} = \text{Bin}(N_{\text{shots}}, p_{k,+}), \quad n_{k,-} = N_{\text{shots}} - n_{k,+}. \quad (12)$$

The measured Stokes parameters are then estimated from the counts as:

$$\hat{S}_k = \frac{n_{k,+} - n_{k,-}}{N_{\text{shots}}}. \quad (13)$$

Thus, for fixed N_{shots} , the raw counts $\mathbf{n}$ and estimated Stokes vector $\hat{\mathbf{S}}$ carry equivalent information about the state. This distinction is nevertheless important for the QST methods considered here: maximum-likelihood estimation operates directly on measurement counts, whereas Stokes reconstruction and the neural network use Stokes values as input.

3. Synthetic Noise Channels

Beyond the statistical uncertainty introduced by finite shots, state preparation & measurement (SPAM) errors emerge by interactions of a system with the experimental environment. To mimic these effects, three phenomenological noise processes are incorporated into the simulated measurement pipeline:

1. Preparation/control: Additive Gaussian Noise Imperfections in state preparation & measurement, such as thermal fluctuations and miscalibrated basis changes, are modelled by perturbing each ideal expectation value from Eq.(10) by an amount drawn from the Gaussian distribution, with zero mean and variance σ^2 , prior to sampling measurement counts:

$$\tilde{S}_k = S_k + \epsilon_k, \quad \epsilon_k \sim \mathcal{N}(0, \sigma^2). \quad (14)$$

Each $\tilde{S}_k$ is clipped to $[-1, 1]$ to remain within the physically valid range, and the identity component $\tilde{S}_0 = 0$ is preserved to maintain $\text{Tr}(\rho) = 1$. The parameter σ controls the noise strength of this model [9].

2. Projection: Finite Shot Noise Measurement uncertainty is modelled through the finite-shot binomial sampling process described above. The number of shots N_{shots} therefore controls the noise strength, where fewer shots indicate larger uncertainty.

3. Readout: Pepper Noise Readout failures and missing measurements are modelled by replacing a random fraction f of Pauli measurement outcomes with uninformative counts drawn from Eq.(12) with $p_{k,+} = p_{k,-} = 0.5$. This corresponds to a complete loss of information for the affected measurement by setting the expectation value to $\hat{S} = 0$, inspired by the pepper noise model in [40].

Altogether, this procedure produces a dataset of ideal target density matrices ρ and their corresponding noisy simulated measurement data. The noise models are not exhaustive, but they provide a general and physically motivated synthetic framework for benchmarking the robustness of the different QST methods under identical conditions in response to research question (ii). Their breadth and flexibility also means they can be refined for a specific experimental setting, as demonstrated in the methodology through the trapped-ion-inspired error budget used to address research question (iii).

B. QST Approaches

Throughout, reconstruction accuracy is quantified by the fidelity between the QST-estimated quantum state $\hat{\rho}$ and the true state ρ :

$$F(\rho, \hat{\rho}) = \left(\text{Tr} \sqrt{\sqrt{\rho} \hat{\rho} \sqrt{\rho}} \right)^2 \quad (15)$$

It satisfies $F \in [0, 1]$ with $F = 1$ indicating perfect reconstruction. For pure states this reduces to $F = \langle \psi | \hat{\rho} | \psi \rangle$, which is directly interpretable as the overlap between the true and reconstructed states. Fidelity is averaged over independent runs on all states in the test dataset, providing a statistically robust performance estimate.

1. Traditional: Maximum-Likelihood Estimation

Maximum-likelihood estimation (MLE) is a statistical QST approach that finds the physical density matrix that maximises the likelihood of obtaining the observed measurement data. Given the simulated counts $n_{k,\pm}$ of each measurement outcome, with a probability $p_{k,\pm}$ associated with the POVM element $E_k^\pm$, the log-likelihood function is:

$$\ln \mathcal{L}(\rho) = \sum_{k,\pm} n_{k,\pm} \ln(p_{k,\pm}) \quad (16)$$

The iterative $R\rho R$ algorithm [41] is adopted to maximise this function subject to the constraints of a physically valid density matrix (Section 2 A). The iteration is initialised in the maximally mixed state $\rho^{(0)} = I/d$ which provides a physical and unbiased starting point. At each iteration t , the update rule is:

$$R^{(t)} = \sum_{k,\pm} \frac{n_{k,\pm}}{p_{k,\pm}(\rho^{(t)})} E_k^\pm, \quad \rho^{(t+1)} = \frac{R^{(t)} \rho^{(t)} R^{(t)}}{\text{Tr}[R^{(t)} \rho^{(t)} R^{(t)}]}. \quad (17)$$

The $R\rho R$ structure preserves PSD at every iteration since R is a positive operator, and normalisation ensures unit trace. Hermiticity is maintained by explicitly symmetrising at each step.

Convergence is assessed by the Frobenius norm [42] between successive iterates

$$\Delta^{(t)} = \frac{\|\rho^{(t+1)} - \rho^{(t)}\|_F}{\|\rho^{(t)}\|_F} < \tau, \quad (18)$$

with tolerance $\tau = 10^{-8}$ chosen to provide a numerical threshold at which successive iterates differ only negligibly [43]. Iteration stops once $\Delta^{(t)} < 10^{-8}$, or once the maximum iteration count is reached. For each input count vector $\mathbf{n} = \{n_{k,+}, n_{k,-}\}$, the final MLE estimate $\hat{\rho}_{\text{MLE}}$ is obtained by applying this iterative procedure.

The maximum iteration count was determined empirically via a convergence benchmark, evaluating mean QST fidelity and per-state inference time over a subset of 500 two-qubit states across five independent runs under fixed noise conditions ($\sigma = 0.10$, $N_{\text{shots}} = 100$, $f = 0.05$). The results are shown in Figure 2.

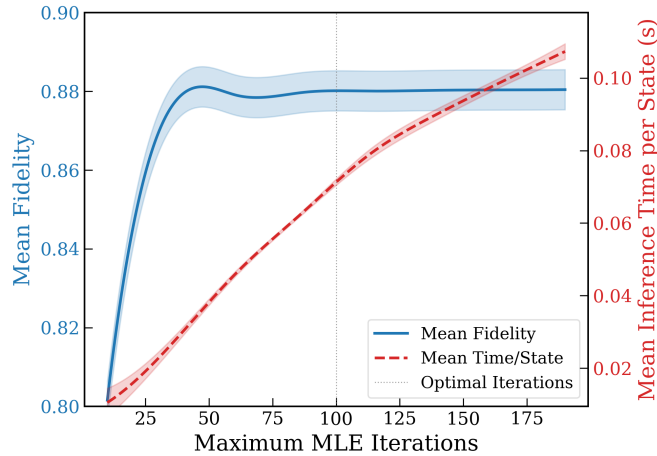


FIG. 2: Convergence of the mean fidelity and mean per-state inference time with maximum iteration count, evaluated over 5 runs on 500 two-qubit states. Error regions represent one standard deviation.

Mean reconstruction fidelity increases rapidly at low iteration number, from 0.802 ± 0.005 at 10 iterations to 0.880 ± 0.005 at 100 iterations, and then plateaus, changing by less than 0.001 between 100 and 200 iterations. In contrast, the mean per-state inference time increased approximately linearly with iteration count, from 6.8ms at 10 iterations to 125.8ms at 200 iterations. When datasets of the orders of 10^3 states are considered, this rapidly increases computational cost. A ceiling of 100 iterations represents a practical compromise.

MLE’s nonlinear objective function and its ability to produce a physical estimate without requiring projection is expected to make it a more robust approach, although has its own limitations in handling noisy data, particularly that it might converge to a non-ideal estimate if the prepared state has been perturbed. Additionally, the standard initialisation in the maximally mixed state may introduce a level of state-dependence when reconstructing states from a diverse pure ensemble.

2. Traditional: Stokes Reconstruction

Stokes reconstruction is the most direct quantum state estimation method, in which the density matrix is reconstructed by substituting the estimated Stokes parameters $\hat{S}_k$ into the inverse Pauli expansion:

$$\hat{\rho} = \frac{1}{d} \sum_{k=0}^{4^n-1} \hat{S}_k P_k. \quad (19)$$

This is a linear, non-iterative estimate with no optimisation procedure or free fit parameters. Unlike MLE, this means that Stokes reconstruction is not guaranteed to estimate physically valid density matrix under the noise constraints; in particular, it may violate positivity. Physicality is enforced by projecting the raw estimate onto the set of valid density matrices such that the final Stokes-reconstructed state is [44]:

$$\hat{\rho}_{\text{Stokes}} = \frac{\hat{\rho}^\dagger \hat{\rho}}{\text{Tr}(\hat{\rho}^\dagger \hat{\rho})}. \quad (20)$$

The main advantage of Stokes reconstruction over MLE is computational efficiency, since its linear inversion does not require iterative optimisation. For the same reason, however, its main limitation is sensitivity to noise: the reconstruction is driven directly by the measured expectation values, so errors are transferred directly into the estimated density matrix. This motivates the use of deep learning as a possible route to solving the limitations of both traditional methods.

3. Deep Learning: MLP Neural Network

As seen in section 2 C, a neural network (NN) contains a plethora of hyperparameters and hence modelling decisions that are required when developing an architecture for a given task. The NN estimator developed in this work is a supervised multilayer perceptron trained to map noisy measurement data to the vectorised density matrix of the corresponding quantum state. Its development proceeded in stages. First, five candidate architectures were tested on a representative noisy two-qubit dataset in order to establish a suitable baseline layer structure. The hidden layer structure of 256–256–128 neurons was selected on the basis of reconstruction fidelity and training stability, with each layer scaling as a factor of the input and output layers. This provides sufficient parametric capacity to learn the nonlinear mapping across the system sizes considered here while remaining computationally tractable.

Bayesian optimisation in Keras refined the learning rate (1.56×10^{-3}) and regularisation strengths (0.2 dropout) for the final model [45]. Dropout regularisation reduces overfitting by randomly deactivating a fraction of hidden neurons during training, preventing the network from relying too heavily on any one subset of features or memorising noise-specific structure, and thereby improving generalisation to unseen data. The model comparisons and optimisation methods are reported in Appendix B.

To visualise the effect of dropout regularisation, the standard and regularised models were trained using a 0.9/0.1 training-validation split on 10000 two-qubit states. This choice preserves the majority of the dataset for weight optimisation while reserving a sufficient subset to monitor overfitting during training. The resulting loss curves are shown in Figure 3.

The standard model in Figure 3a exhibits a persistent gap between the training and validation losses, indicating mild overfitting, whereas the dropout model in Figure 3b converges with the two curves closely aligned. The dropout architecture was therefore selected for all subsequent analysis. The optimal epoch number was determined by early stopping, with training halted once the validation loss had failed to improve for 10 consecutive epochs.

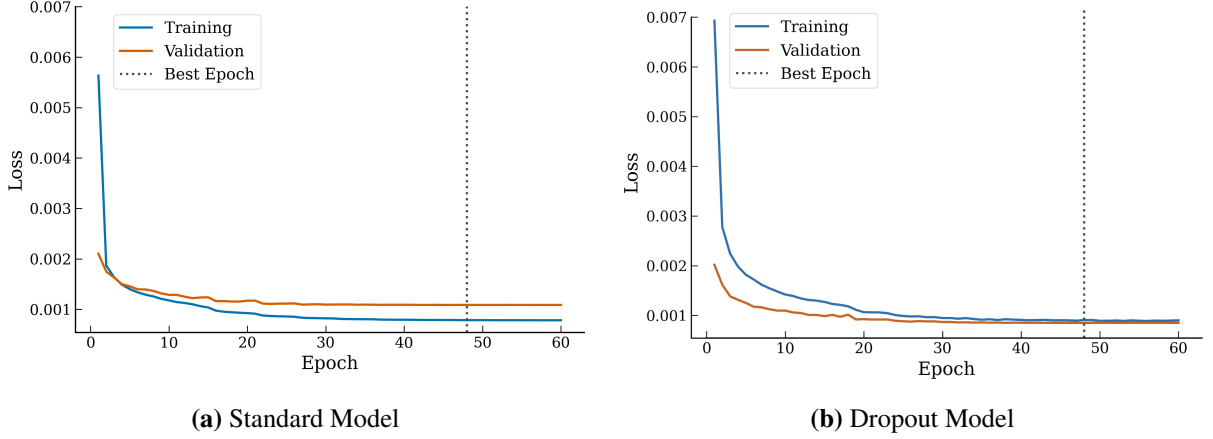


FIG. 3: Training and validation loss curves for: (a) the standard NN architecture, and (b) when a dropout of 0.2 is applied. Dropout reduces overfitting by ensuring the converged validation loss does not exceed that of the training loss. The optimal epoch number for each model is shown by a dashed line.

The final NN architecture is shown schematically in Figure 15. The ReLU activation function, $\phi(x) = \max(0, x)$ is adopted for computational efficiency, resistance to the vanishing gradient problem, and well-established performance in regression and classification tasks [46]. The vectorised output is reshaped into $\hat{\rho} \in \mathbb{C}^{d \times d}$ - since it is not necessarily certain that a valid density matrix is predicted, physicality is enforced to the output using Eq. (19) prior to evaluation of the final $\hat{\rho}_{\text{NN}}$. This reasoning is discussed further in the following interpretability analysis.

The network parameters are optimised using the Adam optimiser, employed here as a standard and effective gradient-based method for supervised learning [9]. Training proceeds by minimising the element-wise mean squared error (MSE) loss function between the predicted and target density matrices [9], derived as:

$$\mathcal{L}_{\text{MSE}}(\hat{\rho}, \rho) = \frac{\sum_{i=1}^{2d^2} \sum_{j=1}^{2d^2} |\hat{\rho}_{ij} - \rho_{ij}|^2}{2d^2}. \quad (21)$$

Training is performed with a mini-batch size of 128, which provides a practical compromise between gradient stability and computational efficiency. The training dataset was determined by monitoring loss convergence with increasing sample size, reported in Appendix B, so that the dataset was large enough to stabilise performance without unnecessary cost.

Training and test datasets were generated with independent random seeds. Each state is produced from the effective unitary circuit with independently sampled depth and gate parameters, ensuring they are unique and statistically independent both within and between the two sets. Once trained, the NN performs quantum state estimation by a single forward pass from the noisy measurement vector to the predicted density matrix.

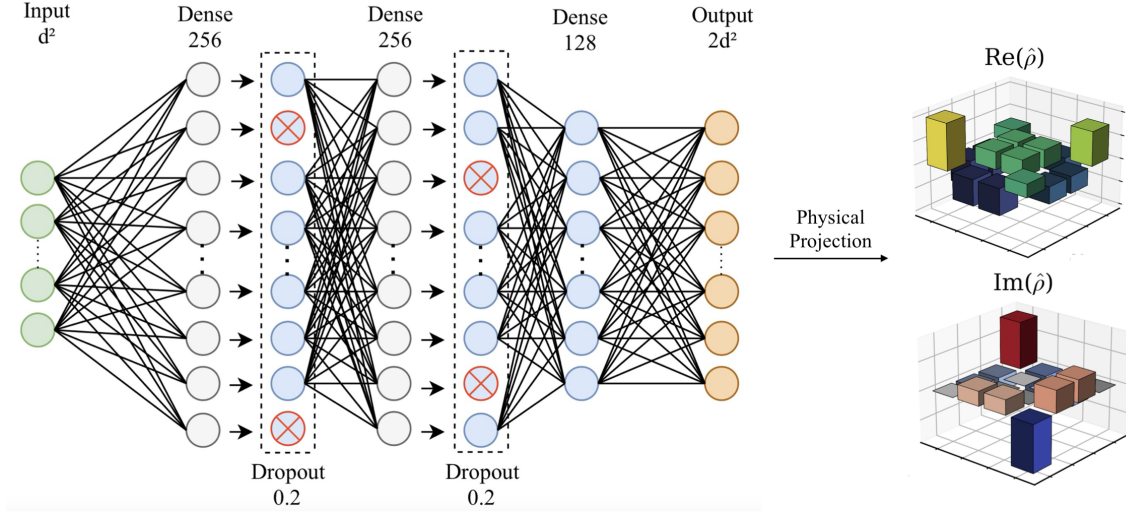


FIG. 4: Schematic of the MLP architecture for deep learning-enhanced quantum state estimation. The noisy stokes vector $\hat{S} \in \mathbb{R}^{d^2}$ is fed into three dense hidden layers of 256, 256, and 128 neurons with ReLU activations. Dropout regularisation (rate 0.2) is applied after each of the first two hidden layers, with randomly deactivated neurons clearly indicated. The linear output layer predicts a $2d^2$ -dimensional real vector encoding the real and imaginary components of the density matrix separately. A projection is then applied to enforce physicality in the reshaped density matrix $\hat{\rho} \in \mathbb{C}^{d \times d}$. The output shown is the reconstruction of the two-qubit example of Figure 1 under representative noise conditions.

C. Benchmarking & Interpretability

1. Abstract Two-Qubit System

For a two-qubit system, the computational basis extends to $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ such that the state is represented by:

$$|\psi\rangle = \alpha_{00} |00\rangle + \alpha_{01} |01\rangle + \alpha_{10} |10\rangle + \alpha_{11} |11\rangle, \quad (22)$$

corresponding to a $\rho \in \mathbb{C}^{4 \times 4}$ density matrix as visualised in Figure 1. The three QST methods are first evaluated in this setting free from external noise sources beyond the intrinsic projection noise of finite quantum measurement; a fixed $N_{\text{shots}} = 100$ is chosen to be sufficiently large and computationally efficient for this purpose. The fidelity distribution over $N = 5000$ test states, chosen as a sufficiently large sample for stable and representative ensemble statistics [9], is estimated via kernel density estimation (KDE) [47], which provides a smooth, non-parametric representation of the per-state fidelities. This gives a rich characterisation of each method's performance, revealing fidelity average and consistency across all states. These results address whether DL-based QST is a feasible and competitive method in an abstract two-qubit setting.

Having established feasibility, external noise is introduced to benchmark performance under conditions representative of a physical system. Each noise source is swept independently with the remaining two held at their noiseless values, isolating the contribution of each to reconstruction performance. Since finite measurement noise is always present, $N_{\text{shots}} = 10000$ is fixed to render its effect negligible when other noise sources are being investigated. The noise ranges are summarised in Table I.

TABLE I: Noise sweep values used in the QST simulations.

Noise process	Parameter	Values
Preparation/Control	σ	{0.00, 0.06, 0.12, 0.18, 0.24, 0.30}
Projection	N_{shots}	{10, 50, 100, 500, 1000, 5000}
Readout	f	{0.00, 0.05, 0.10, 0.15, 0.20, 0.25}

For each noise value, the mean fidelity is computed over five independent runs on the test dataset. Performance is benchmarked by mean fidelity as a function of noise strength, characterising their relative robustness to SPAM and projection errors. These results are presented in Section 4 B.

2. Interpretability of Neural Network

Neural networks are black boxes - they take an input and predict an output, but how are the hidden layers performing this transformation? At this stage, this insight could help to understand not just the fact that deep learning might outperform traditional methods but why, by understanding what it is doing differently. To investigate this, a physical interpretability analysis is performed on a representative noisy two-qubit dataset.

This is carried out by measuring the alignment between the learned weight vectors of the hidden layer neurons and the physical two-outcome measurement projectors (POVM elements) used to obtain tomographic measurements. For each non-identity Pauli observable P_k , the corresponding projectors,

$$E_k^\pm = \frac{1}{2}(I \pm P_k) \quad (23)$$

are expanded in the two-qubit Pauli basis, giving a matrix compiled of $2(4^n - 1) = 30$ individual projector coefficient vectors in $\mathbb{R}^{4^n}$ for the two-qubit example. The trainable weight vectors of each dense hidden layer are extracted and compared to this projector matrix. Since the first hidden layer acts directly on the $d^2 = 4^n$ -dimensional Stokes input, its neuron weights can be compared to the projector coefficients directly. For deeper hidden layers, whose weight vectors lie in higher-dimensional spaces, principal component analysis (PCA) [48] is employed to reduce the dimensionality while retaining representative information.

Directional alignment is quantified using cosine similarity. For a neuron weight vector $\mathbf{w}_i$ and projector coefficient vector $\mathbf{p}_j$, the cosine similarity is defined as [49]:

$$\cos(\mathbf{w}_i, \mathbf{p}_j) = \frac{\mathbf{w}_i \cdot \mathbf{p}_j}{\|\mathbf{w}_i\| \|\mathbf{p}_j\|}. \quad (24)$$

This takes a value between -1 and 1 , with positive and negative values corresponding to parallel and anti-parallel orientation respectively. The absolute value is taken so that alignment is normalised between 0 and 1 . For each neuron, the projector with the largest absolute cosine similarity is identified. The distributions of best-matched cosine similarity for neurons in the three hidden layers are shown in Figure 5.

In all cases, the similarity values are broad and centred at only moderate alignment, with median absolute cosine similarities of approximately 0.51 , 0.54 , and 0.54 . No layer exhibits a

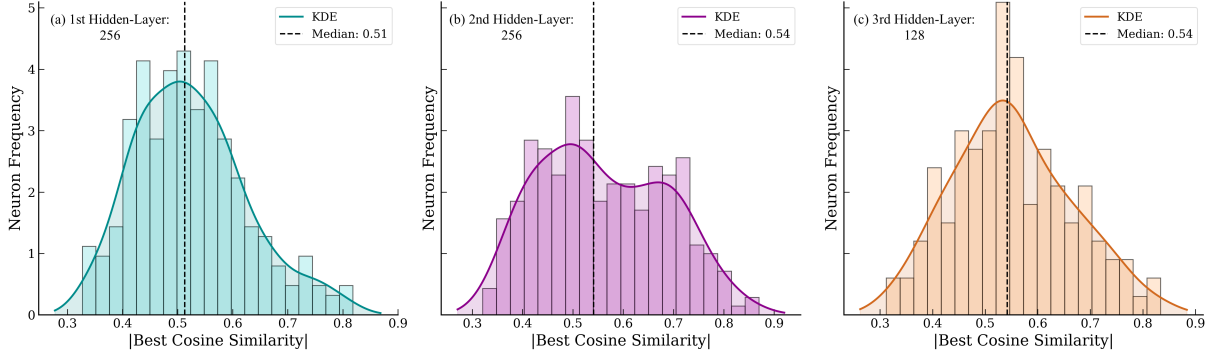


FIG. 5: Distributions of the best absolute cosine similarity between hidden-layer neuron weight vectors and two-qubit Pauli-projector coefficient vectors for the three dense hidden layers of the final dropout model. Dashed lines indicate the median similarity in each layer.

sharp concentration near unity, and therefore no interpretable projector-like measurement structure is recovered at any point in the network. This suggests that the network does not internalise tomography in the same form as Stokes reconstruction or MLE. Instead, somewhat as expected, it learns its own nonlinear representation for mapping noisy measurements to the target density matrix, tuned for denoising and minimising loss. For this reason, it cannot be certain that the network predicts a physically valid output, motivating the enforced projection.

The distribution of best-matched projector labels, shown in Appendix B, provides a secondary view of this behaviour. Some preference for particular projector families is visible, especially in the middle hidden layer, indicating that the learned representation is not completely structureless. However, because these preferences are accompanied by only moderate cosine alignment, they should not be overinterpreted as evidence that the network has recovered a physically meaningful structure. Rather, they suggest that certain measurement directions are reused more strongly than others within a broader distributed representation.

3. Physical Application: Trapped-Ion Processor

The independent noise sweeps establish the robustness of each QST method across a broad parameter space for an abstract two-qubit system, using phenomenological noise models chosen for their versatile interpretability. The final research question concerns scalability with system size. To address this in a physically motivated setting, the abstract framework is refined into a single realistic operating point inspired by the dominant reported error sources of a $^{40}\text{Ca}^+$ trapped-ion processor [50]. This does not alter the methodological focus of the study; rather, it provides a grounded example of how the general framework can be specialised to a realistic platform for which scalable quantum state estimation is a central challenge.

In a $^{40}\text{Ca}^+$ trapped-ion processor, qubits are encoded in two long-lived internal electronic states of individual ions held in the harmonic potential of a linear Paul trap. The computational basis is identified as $|S_{1/2}\rangle \equiv |0\rangle$ and $|D_{3/2}\rangle \equiv |1\rangle$ where $S_{1/2}$ is the electronic ground state and $D_{5/2}$ is a metastable excited state. The ions are initialised into the ground state by optical pumping and manipulated by coherent laser pulses that implement single-qubit rotations and entangling gates through the shared motional modes of the ion chain. Readout is performed

by state-dependent fluorescence on a CCD camera: ions in the bright $|S\rangle$ state scatter photons under resonant illumination, whereas ions in the dark $|D\rangle$ state do not. Repetition of this binary outcome preparation–measurement cycle yields the tomographic counts used for state reconstruction. Calcium ions serve as a particularly well-studied implementation whose long-lived electronic states are well suited for qubit encoding.

While the same overall QST pipeline is retained, the abstract noise models are refined into trapped ion-inspired channels. The additive Gaussian perturbation is replaced by two explicit state-level channels: a coherent over-rotation of width σ_θ representing imperfect laser control and ion-addressing error, and a local phase-flip channel of strength p_ϕ representing laser detuning that suppresses coherence. Likewise, the pepper-readout model is replaced by a bit-flip readout channel, in which a fraction f_{flip} of measurement outcomes is misassigned between the bright and dark states, representing fluorescence misclassification and cross-talk. Finite-shot projection noise is retained since it is intrinsic to the repeated binary measurement process. The mapping from the abstract framework to the specific channels used in the scalability benchmark is summarised in Table II. Full channel definitions, implemented via definitions provided in [1], together with the relevant energy-level structure and further details of qubit encoding in the trapped-ion processor, are presented in Appendix D.

TABLE II: Refinement of the abstract noise models into the trapped-ion-inspired channels used in the scalability benchmark. Values best reflect the relative contributions in the cited error budget.

Abstract model	Refined channel	Physical interpretation	Value
Additive Gaussian	Coherent overrotation	Addressing error	$\sigma_\theta = 0.08$
	Local phase-flip	Laser frequency noise	$p_\phi = 0.06$
Projection	–	Finite measurement cycles	$N_{\text{shots}} = 100$
Pepper readout	Bit-flip readout error	Fluorescence misclassification	$f_{\text{flip}} = 0.02$

Under this combined noise setting, each QST method is evaluated on test datasets of 1000 states for $n = \{2, 3, 4\}$ qubit systems. This reduced dataset maintains computational feasibility as the Hilbert-space dimension grows exponentially, while still providing a sufficiently large and diverse sample for stable average performance metrics. For each system size, the mean fidelity and the total inference time required to reconstruct the full test set are averaged over five independent runs, thereby benchmarking both reconstruction accuracy and computational cost as functions of qubit number. Building on the research inspiring this work [50, 51, 52], this final comparison tests whether DL offers a more scalable route to quantum state estimation than traditional Stokes and MLE under realistic noise conditions.

4. RESULTS & DISCUSSION

A. Feasibility of Deep Learning for Quantum State Tomography

Figure 6 shows the fidelity distributions of the three QST methods applied to the two-qubit test dataset under finite projection noise with fixed $N_{\text{shots}} = 100$. All three methods achieve

mean fidelities $\bar{F} > 0.96$, confirming that each is capable of reconstructing the density matrix from simulated measurement data in this abstract setting.

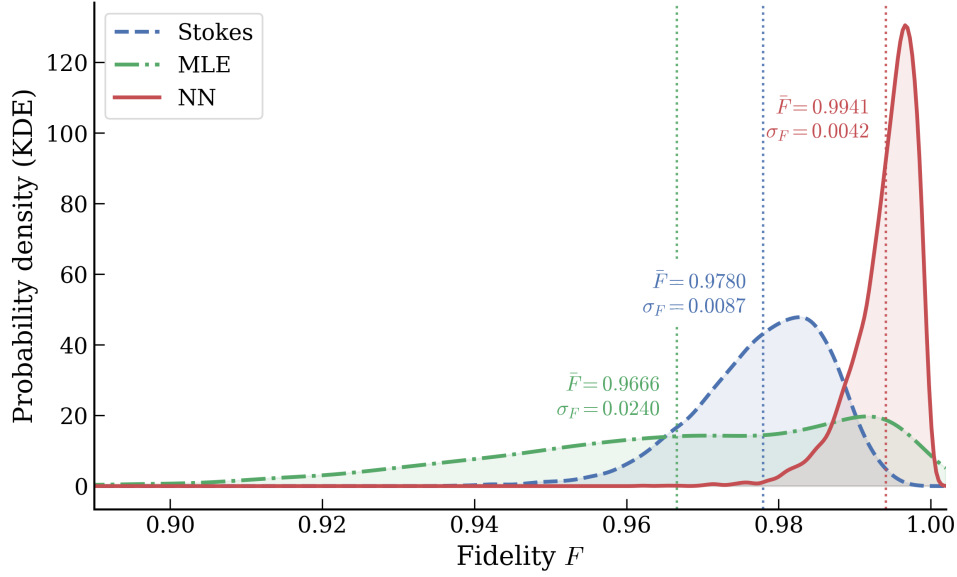


FIG. 6: Fidelity distributions of Stokes reconstruction, MLE, and the NN estimator applied to the two-qubit dataset ($N = 5000$ states) under finite projection noise only ($N_{\text{shots}} = 100$). Distributions are estimated via Kernel Density Estimation (KDE) over per-state fidelities, with dotted lines indicating the mean fidelity $\bar{F}$ for each method, annotated alongside the standard deviation σ_F quantifying the spread.

The Stokes distribution ($\bar{F} = 0.978$, $\sigma_F = 0.009$) is unimodal and moderately broad. This is consistent with the nature of linear inversion: binomial shot noise in the measured expectation values is propagated directly into the reconstructed density matrix $\hat{\rho}$, producing a relatively uniform spread of fidelities across the test ensemble about the mean with no state-dependence. No optimisation or denoising step is involved, so the method is simple and stable, but its performance is limited by the statistical quality of the measured data. Its relatively strong performance here should be interpreted in light of the projection of the raw estimate onto the physical state space which ensures validity and partially compensates for shot noise.

The MLE distribution ($\bar{F} = 0.966$, $\sigma_F = 0.024$) is broader and exhibits a more pronounced low-fidelity tail. This indicates that MLE is more variable across the test ensemble in this finite-shot regime. Unlike Stokes reconstruction, MLE depends on iterative likelihood optimisation from noisy count data, so its performance can be more sensitive to statistical fluctuations and to the structure of the likelihood landscape for individual states. In particular, some states appear to be reconstructed less reliably than others, leading to the asymmetric spread. While the detailed origin of this state dependence is not resolved here, it could arise from varying levels of entanglement induced by the random sampling of circuit depth and gate angles in the dataset generation procedure; highly entangled states, which dominate the ensemble, are reconstructed reliably, while the low-entanglement minority populates the left tail and lowers the mean fidelity. The result suggests that, at $N_{\text{shots}} = 100$, the iterative optimisation does not consistently outperform direct inversion across the full ensemble, implying Stokes reconstruction degrades more uniformly than MLE regardless of the underlying states.

The NN distribution ($\bar{F} = 0.994$, $\sigma_F = 0.004$) is sharply peaked near unity, with both the highest mean fidelity and the smallest spread. This provides strong evidence that supervised deep learning can perform quantum state estimation accurately and consistently in this setting. Trained on 10^4 noisy Stokes vectors paired with their true density matrices, the network has learned a regularised nonlinear mapping from tomographic measurement data to the reconstructed state that appears to substantially suppress the finite-shot fluctuations present in the input. Its narrow distribution indicates that this mapping generalises well across the unseen test ensemble rather than performing well only on a restricted subset of states. In contrast to Stokes reconstruction, the NN is not limited to a fixed linear inversion and hence can distinguish between signal and noise, and unlike MLE it does not rely on per-state iterative optimisation at inference. The combination of nonlinear hidden-layer ReLU activations, supervised training against the true target states, and dropout regularisation therefore appears to improve both accuracy and per-state consistency.

These results directly address research question (i): deep learning is not only a feasible approach to quantum state estimation, but in this finite-shot two-qubit setting it outperforms both Stokes reconstruction and MLE across the test ensemble. The KDE distributions reveal an ordering of performance in this initial benchmark. Projected Stokes reconstruction provides a strong and relatively stable baseline, MLE exhibits greater variability that reduces its mean performance, and the NN achieves the best overall combination of accuracy and consistency. Its advantage is therefore not confined to a small subset of favourable states, but reflects systematically stronger generalisation across the full dataset. The following section extends this analysis by introducing additional simulated physical noise channels and benchmarking the QST methods across their corresponding parameter ranges.

B. Robustness to Noise in an Abstract Physical System

Having established the NN’s advantage under a fixed finite noise setting, the three QST methods are next evaluated across the three independent noise regimes for the two-qubit ensemble. In all cases, the NN maintains the highest mean reconstruction fidelity, indicating greater robustness to physically motivated SPAM errors and projection noise than either Stokes reconstruction or MLE. The results are shown in Figure 7.

The overall advantage of the NN reflects the different ways in which the three estimators process noisy input data. Stokes reconstruction is a direct linear inversion, so noise in the measured expectation values is transferred immediately into the reconstructed density matrix prior to enforced physical projection. MLE enforces physicality via its iterative algorithm and maximises the likelihood of the observed counts, which can make it more stable than direct inversion in some regimes, but its performance remains limited by the informativeness of the noisy data to the underlying state and the shape of the resulting likelihood landscape. The NN, by contrast, learns a supervised nonlinear mapping from noisy measurements to density matrices by systematically adjusting its parameters to minimise the loss between the output and target from a large training ensemble generated under matched noise conditions.

In particular, the ReLU activation function introduces nonlinearity, allowing the hidden-layer neurons to build increasingly rich features from correlations between many observables

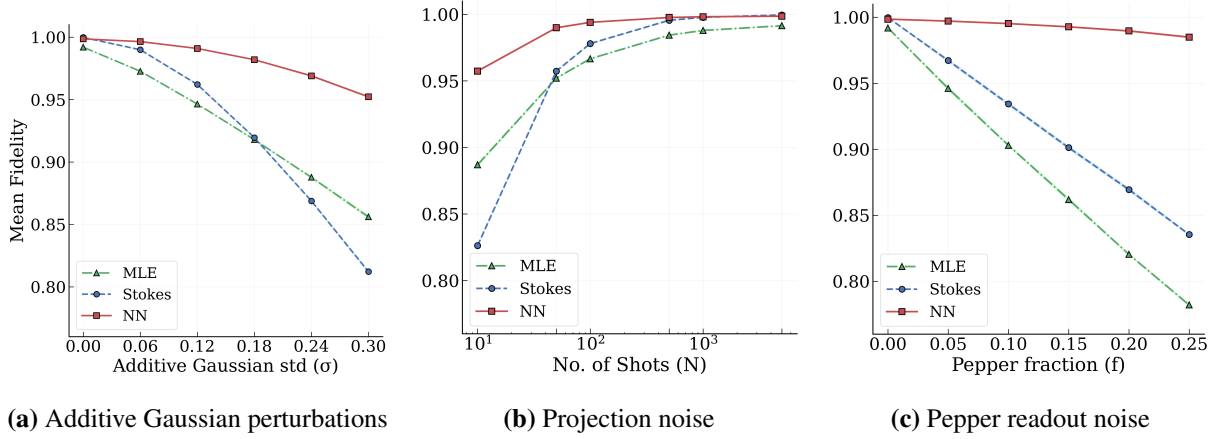


FIG. 7: Mean reconstruction fidelity as a function of noise level for the three QST methods under (a) additive Gaussian preparation noise, (b) finite-shot projection noise, and (c) pepper readout noise. These curves illustrate the robustness of each method to different error sources in the measurement pipeline. Shaded error regions, where visible, represent one standard deviation about the mean.

rather than treating each noisy measurement independently. This enables the network to learn which patterns in the input are consistently associated with genuine state structure and which are more likely to arise from statistical or experimental noise, so that the final prediction acts as a learned denoising map rather than a direct inversion of the corrupted data. Dropout regularisation further discourages the model from relying too strongly on any one subset of features or memorising noise-specific structure, improving its ability to generalise to unseen states and maintain high fidelity across a broad range of noise conditions..

Figure 7a shows the effect of additive Gaussian perturbations to the Stokes parameters, representing a phenomenological model of preparation and control noise. All three methods achieve near-perfect fidelity at $\sigma = 0$, constrained only by the near-negligible projection noise, and degrade monotonically with increasing noise. The NN degrades most slowly, maintaining $F > 0.95$ across the full range, whereas Stokes reconstruction and MLE fall to approximately $F \approx 0.81$ and $F \approx 0.86$, respectively, at $\sigma = 0.30$. In the low-noise regime, Stokes reconstruction remains slightly stronger than MLE, MLE exhibits greater robustness to large perturbations. This is consistent with the fact that MLE incorporates an explicit physicality constraint and nonlinear likelihood-based optimisation, whereas Stokes reconstruction transfers the corrupted expectation values directly into the estimate. The NN’s much flatter degradation suggests that its learned nonlinear mapping is able to partially compensate for these perturbations across the full test ensemble.

The projection noise results in Figure 7b show a qualitatively similar pattern. Finite-shot noise appears as systematic fluctuation in the input measurement vector, and the NN’s layered nonlinear architecture allows it to learn correlations between observables that distinguish this fluctuation from genuine state structure. In this sense, the network acts as a learned denoiser, whereas Stokes reconstruction transfers the noise directly and MLE must optimise against noisy counts on a state-by-state basis. For low shots, where projection noise is strongest, MLE initially outperforms Stokes reconstruction, with fidelities of approximately $F \approx 0.89$ and $F \approx 0.83$ at $N_{\text{shots}} = 10$, while the NN maintains $F \approx 0.96$. As the shot number increases, however, Stokes

reconstruction overtakes MLE and converges more rapidly toward unit fidelity. This suggests that MLE is somewhat more stable in the extremely data-limited regime, where its nonlinear iterative optimisation regularises the estimate, but that it becomes less advantageous once modestly informative count statistics are available. As discussed in the previous section, this could reflect certain types of states being reconstructed poorly causing convergence to non-ideal estimates even when the number of shots and iterations are sufficient to optimise its performance. All three methods approach near-perfect reconstruction in the high-shot limit, as expected when the measured counts converge to the true probabilities in the absence of additional noise. The NN nevertheless maintains the strongest performance throughout, which is particularly relevant for resource-constrained experiments where the number of measurements is limited.

The pepper readout noise results in Figure 7c reveal the most striking divergence between methods. The NN fidelity again degrades only marginally, maintaining $F \approx 0.99$ even at $f = 0.25$, while both MLE and Stokes show a steep linear decline. Notably, Stokes consistently outperforms MLE across the whole range - reaching $F \approx 0.83$ while MLE diverges to $F \approx 0.78$. This observation lies in how the two traditional methods interpret missing measurements. When a fraction of estimated expectation values $\hat{S}_k$ are set to zero, they simply contribute nothing to the linear Stokes inversion - the fidelity degrades appropriately because the missing terms are effectively ignored. MLE, however, treats the uninformative counts drawn from $p_{k,\pm} = 0.5$ as statistical evidence that $\hat{S}_k = 0$, directing the estimate towards a maximally mixed state in the affected dimensions. Evidently, this causes it to converge to a systematically worse estimate than Stokes, which inadvertently handles the missing data more robustly by treating the zeroed components as absent rather than informative. The NN's near-immunity to pepper noise further supports its clear advantage: having been trained on data with missing measurements, it has learned to recognise the signature of uninformative observables and discount them appropriately while maintaining high fidelity.

Taken together, these results address research question (ii) by testing whether DL-based quantum state estimation remains effective in a noisy abstract physical setting. The answer appears affirmative - the NN remains systematically more robust than both traditional methods across all three regimes. The noise models used here are intentionally phenomenological rather than platform-specific, but they capture a physically motivated error budget: preparation/control perturbation such as overrotations or thermal fluctuations, finite measurement statistics, and readout corruption are ubiquitous in real-world experiments. The NN's robustness across all three regimes suggests that the learned estimator has acquired a generalised noise-robust representation of the mapping from measurements to quantum states, supporting the use of DL as a practically viable alternative to traditional QST methods provided that training data can be generated under representative noise conditions.

A potential statistical limitation emerges by averaging each noise point over only five Monte Carlo runs. Nevertheless, the trends are qualitatively clear, and this choice was necessary for computational feasibility, particularly in view of the larger-system scaling analysis that follows. The small spread between runs, reflected in the narrow error regions in Figure 7, suggests that the reported averages are sufficiently stable for the purposes of this study. Future work could strengthen the statistical grounding of these results by increasing the number of runs and applying formal significance tests to quantify the performance improvement.

Experimental validation on real tomographic datasets would be a valuable next step in de-

termining the extent to which the present simulation-based observations transfer to practical quantum hardware. Using matched training and test distributions is a potential assumption, and future work could therefore strengthen these conclusions by testing robustness to distribution shift and drifting noise, and extending the comparison to larger, more structured and different classes of quantum states such as those highlighted in Appendix C.

C. Scalability with System Size

The noise-sweep results characterise each method’s robustness to individual error sources in isolation. A real physical system, however, experiences these imperfections simultaneously, so the remaining question is how fidelity and computational cost evolve with system size under a fixed, physically motivated noise condition. Figure 8 presents the mean reconstruction fidelity and total inference time over a 1000-state test set as functions of qubit number, $n \in \{2, 3, 4\}$, under the trapped-ion-inspired operating point derived in Section 3 C 3. This benchmark extends the earlier abstract two-qubit analysis by testing whether the observed performance trends persist as the state space grows.

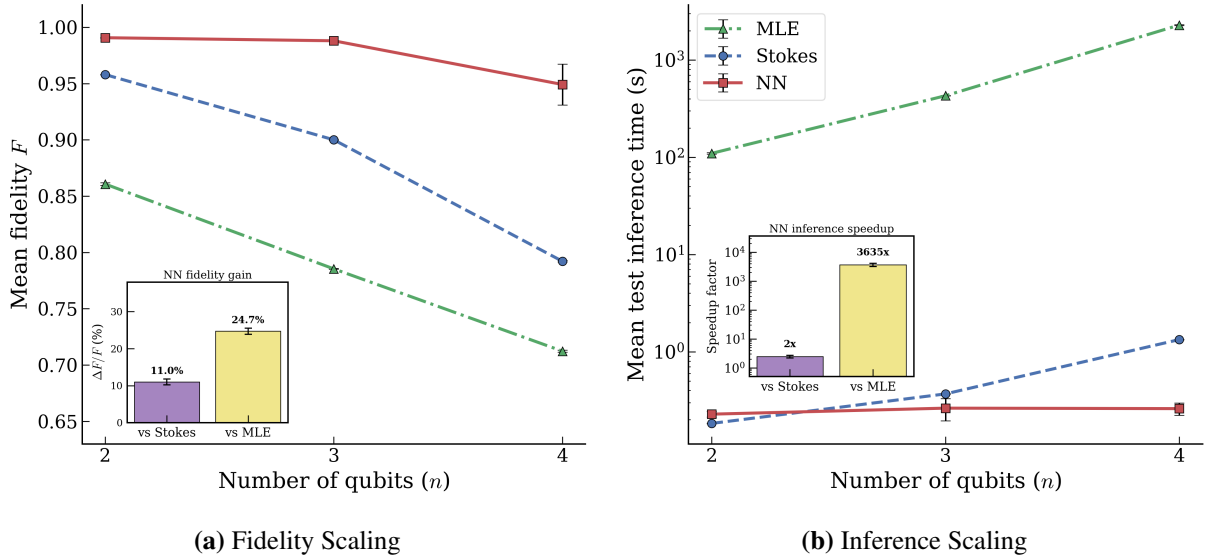


FIG. 8: Scalability of QST performance with qubit number $n \in \{2, 3, 4\}$ under a trapped-ion-inspired noise condition derived from the error budget of a $^{40}\text{Ca}^+$ processor [50]. Results are averaged over 5 independent runs on 1000 test states, with error bars denoting one standard deviation. (a) Mean fidelity $\bar{F}$ as a function of n . The inset shows the mean normalised fidelity improvement $\Delta\bar{F}/\bar{F}$ of the NN over Stokes and MLE over the considered range. (b) Mean inference time over the full test dataset. The inset shows the mean inference speedup factor $\bar{T}_{\text{method}}/\bar{T}_{\text{NN}}$ of the NN over the full n range.

A universal decrease in fidelity and increase in inference time with increasing qubit number is expected and this trend is clearly observed. As n increases, the Hilbert-space dimension grows exponentially with $d = 2^n$, so the number of observables required for full Pauli-basis tomography of the density matrix grows as $d^2 = 4^n$. This rapidly increases the computational cost of processing the full dataset. The control, dephasing, and readout noise channels affect a larger number of measurement parameters, and with the shot budget fixed at $N_{\text{shots}} = 100$

per observable, the reconstruction must combine an exponentially growing number of noisy measurement estimates, making the final state estimate increasingly sensitive to accumulated error.

1. Fidelity Scaling

The NN maintains the highest mean fidelity, degrading from $\bar{F} = 0.994$ at $n = 2$ to $\bar{F} = 0.950$ at $n = 4$. This robustness showcases the network’s capacity to learn a generalised mapping that extends to larger systems after retraining for each system size under the corresponding noise condition. The inset of Figure 8a shows that the mean fidelity gain of the NN is approximately 11% over Stokes reconstruction and 25% over MLE across the full qubit number range considered. This is a central result that demonstrates the ability of the NN to mitigate noise that becomes more pronounced in larger systems. The learning and dropout rates, training dataset size, and epoch number were tuned for $n = 2$, but the fidelity robustness suggests this architecture transfers reasonably well to larger systems. Dedicated hyperparameter tuning for each n could nevertheless improve performance further and is a consideration for future work.

The steeper fidelity drop and larger run-to-run variation at $n = 4$, visible in the pronounced error bar ($\sigma_F = 0.02$), are informative. It suggests that the fixed architecture may be approaching the limit of what it can represent robustly at the largest system size considered. In particular, while the input and output dimensions scale with the Hilbert space dimension, the hidden-layer widths are fixed and tuned in the smaller-system regime. For $n = 4$, the input layer width $d^2 = 256$ matches the first hidden layer width exactly, which may reduce the network’s capacity to compress the input measurement noise in each training run. This does not undermine the main result since the NN still remains clearly superior in mean fidelity, but it does suggest that architecture scaling and qubit-specific hyperparameter tuning could improve the stability of performance for larger systems.

Stokes reconstruction degrades more steeply, from $\bar{F} = 0.958$ at $n = 2$ to $\bar{F} = 0.792$ at $n = 4$. As established, the linear inversion propagates noise directly into $\hat{\rho}$ without compensation, and as the number of noisy observables grows exponentially the accumulated error in the reconstructed density matrix grows accordingly. The enforced physicality projection partially mitigates this by discarding the unphysical component of the noisy estimate, but cannot recover any ideal information from noisy measurements.

MLE shows the steepest fidelity degradation, from $\bar{F} = 0.861$ at $n = 2$ to $\bar{F} = 0.712$ at $n = 4$, and performs worst at every qubit number. In the present implementation, this appears to reflect compounding sensitivity of the iterative $R\rho R$ algorithm to corrupted measurements as dimension increases. The coherent over-rotation and dephasing channels act on the density matrix prior to measurement, so the reconstructed state is increasingly constrained by noisy counts that reflect a quantum state further from the target [53, 54]. The bit-flip readout channel further hinders performance by treating an increasing fraction of misassigned counts as informative evidence of an incorrect outcome. This effect is compounded by the unbiased maximally mixed initialisation, which becomes progressively further from the pure target states. Increasing the iteration budget might therefore improve fidelity, but would further worsen an already

prohibitive inference time. A more effective improvement could be better initialisation, for example from the linear Stokes estimate, or the use of alternative likelihood solvers. The extent of the poor scaling observed here should be interpreted primarily as a limitation of the MLE implementation used in this work rather than of maximum-likelihood tomography in general.

2. Inference Scaling

The inference-time results provide further evidence for the superior scalability of the NN - reconstruction time remains approximately flat over the tested range, staying close to 0.22–0.26s for the full 1000-state dataset. Inference behaves as an effectively fixed-cost forward pass through the trained network, and once trained, the model reconstructs all test states through batched matrix operations rather than state-by-state inversion or iterative optimisation. This explains the weak dependence on system size over the small- n regime considered here. The NN is essentially superior since it can reconstruct the full measurement vector at once, rather than processing observables independently for each state. This allows it to exploit correlations in the data and apply, at inference time, a learned inverse map that has already absorbed much of the complexity of the reconstruction task during training.

MLE, by contrast, becomes dramatically more expensive as system size increases, rising from around 100s at $n = 2$ to more than 40 minutes at $n = 4$, and is by far the most computationally expensive method at every qubit number. This rapid growth is expected. At each iteration, the $R\rho R$ algorithm must evaluate $\text{Tr}(\rho E_k^\pm)$ for all $2(4^n - 1)$ POVM elements and then update the density matrix accordingly, with this process repeated for up to 100 iterations per state. The resulting exponential overhead makes this MLE implementation impractical beyond a few qubits. Improved likelihood-based tomography schemes, such as compressed-sensing and tensor-network methods, have been proposed to reduce the underlying cost by exploiting low-rank or structured states [55]. The results of this research instead promote deep learning as the most superior alternative route to scalable quantum state estimation.

Stokes reconstruction remains much cheaper than MLE, increasing from sub-second runtime at $n = 2$ to around 1.3s for $n = 4$. Its scaling is therefore substantially more favourable than MLE, but still worse than the NN over the tested range because each state must still be reconstructed individually through explicit inversion and subsequent physical projection. As the number of Pauli observables grows exponentially with qubit number, so too does the amount of tomographic data that must be processed for each state, and the inference time scales accordingly. This means despite being comparable or even slightly faster than the NN at $n = 2$, it will quickly become less favourable once large systems are considered.

3. Remarks & Implications

It is important to note that training time is not included in this comparison, and that this is the appropriate choice. In practice, a neural network would be trained once on data representative of a device’s operating error budget and then reused for inference on new measurements. Training is therefore a one-time overhead, whereas inference cost is incurred for every reconstructed state. On this metric, the NN is clearly advantageous: the inset of Figure 8b shows a mean

speedup of approximately 2 times over Stokes and more than 3.5×10^3 times over MLE across the tested range. For larger systems, training would become more demanding because both the input and output dimensions grow rapidly with qubit number, requiring larger network capacity and training data. Even so, this cost remains amortised across all subsequent reconstructions, so the practical advantage of fast inference over traditional methods would still be expected to persist.

Although the final benchmark is grounded in a trapped-ion-inspired setting, the results have broader implications. As controllable quantum processors continue to grow across a range of architectures, including superconducting and photonic platforms [56, 57], scalable state estimation remains a broadly relevant problem. Extending the analysis to other state families and platform-specific noise channels would therefore confirm the generalisability of the synthetic framework developed here. Since this benchmark remains simulation-based, future work could also strengthen these conclusions by validating the estimators on real experimental datasets, and testing the sensitivity of the NN to distribution shift between training and testing noise.

Taken together, Figure 8 identifies the NN as the most scalable of the three approaches considered in this work, directly answering the final research question. It achieves the highest fidelity at every system size while maintaining the most favourable inference-time scaling. Although the present benchmark is restricted to an accessibly small-system regime, the conclusion is physically well motivated: current processors operate at substantially larger qubit numbers [58], making scalable state verification increasingly important. Extending the analysis to more qubits is therefore the natural next step in determining whether the near-flat NN inference time trend and strong fidelity robustness observed here persist beyond the present range, potentially allowing empirical scaling relations to be quantified more rigorously. Nevertheless, the qualitative trends found here already provide strong evidence that deep learning offers a promising route to accurate, computationally efficient, and scalable quantum state estimation.

5. CONCLUSIONS

This research developed and benchmarked a supervised multilayer perceptron neural network (NN) for quantum state tomography, with the aim of determining whether deep learning provides a feasible, noise-robust, and scalable alternative to traditional reconstruction methods. The study was structured around three primary research questions: (i) whether deep learning is a viable approach to quantum state estimation, (ii) whether it remains applicable to noisy physical systems, and (iii) how its performance scales with system size. To address these, the NN was compared against traditional Stokes reconstruction and maximum-likelihood estimation (MLE) in a sequential investigation of increasing physical realism, progressing from an abstract two-qubit benchmark to independent phenomenological noise sweeps and finally to a trapped-ion-inspired scalability test. Across all conditions investigated, the NN consistently achieved the strongest overall performance in reconstruction fidelity, noise-robustness, and computational efficiency.

In the ideal abstract two-qubit setting, the NN achieved the highest mean fidelity and the narrowest fidelity distribution, demonstrating superior accuracy and consistency relative to traditional QST methods across a large and diverse test ensemble. The results showed that the

learned estimator was not only more successful on average, but also substantially less sensitive to state-to-state variation than the traditional methods, answering the first research question affirmatively. Although the MLE results highlighted apparent limitations of the iterative implementation used here, the central result remains that the NN provided the strongest reconstruction performance among the methods tested. The subsequent independent noise sweeps answered research question (ii) by showing that the NN remained the most robust method under three simulated error classes: additive Gaussian perturbations, finite-shot projection noise, and pepper readout corruption. In each case, the fidelity of the NN prediction decreased significantly slower than both Stokes reconstruction and MLE, meaning the relative advantage of employing a NN increased as the noise strength grew. This indicates that the supervised model had learned a nonlinear mapping capable of separating meaningful state structure from measurement corruption more effectively than either linear inversion or iterative likelihood optimisation. Although the noise models used here were phenomenological, they were chosen to represent broad and physically interpretable classes of experimental imperfection. The results therefore support the conclusion that deep learning can provide a practically useful route to quantum state estimation in noisy settings, provided that representative training data can be generated.

The most significant result is the final scalability benchmark. Under a refined trapped-ion-inspired combined noise condition, the NN maintained the highest fidelity at every tested system size while also exhibiting clearly the most favourable, near-constant, inference time scaling. By contrast, Stokes reconstruction degraded steadily and MLE became both less accurate and dramatically more expensive with growing system size. At $n = 4$, for example, MLE required 2300 seconds for 1000 states while the NN completed the same task in under 0.3 seconds - a speedup by a factor of $(8.2 \pm 1.2) \times 10^3$ coupled with a normalised fidelity improvement of $33 \pm 3\%$, maintaining $\bar{F} = 0.95$. This directly answers the final research question and identifies deep learning as the most scalable of the three approaches considered in this work. A natural next step would be to develop the framework into a practical software tool that can be adapted and refined for real experiments, while extending it to larger systems and other types of quantum states.

The significance of these results extends beyond the specific benchmarks considered here. Accurately identifying a quantum state in the presence of noise is a broadly relevant challenge across contemporary quantum computing hardware and other areas of quantum information science, including communication, metrology, and optics. In that broader context, this research provides strong evidence that deep learning offers a promising solution to the limitations of traditional reconstruction methods, especially as systems become larger and noisier, suggesting that learned estimators may play an increasingly important role in the characterisation and verification of future quantum technologies. This reflects the wider bitter lesson that artificial intelligence methods that can learn directly from data generally outperform approaches that rely solely on human knowledge.

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REFERENCES

- [1] M. A. Nielsen, I. Chuang & L. K. Grover, Quantum Computation and Quantum Information, *Am. J. Phys.* **70**(5), 2022. doi:10.1119/1.1463744
- [2] C. Ferrie and R. Blume-Kohout, Maximum likelihood quantum state tomography is inadmissible, arXiv preprint arXiv:1808.01072, 2018. doi:10.48550/arXiv.1808.01072
- [3] D. V. F. James, Measurement of qubits, *Physical Review A* **64**:052312, 2001. doi:10.1103/PhysRevA.64.052312
- [4] S. Lohani et al, Machine Learning assisted quantum state estimation, *Mach. Learn.: Sci. Technol.* **1**:035007, 2020. doi:10.1088/2632-2153/ab9a21
- [5] R. Sutton, Bitter Lesson, www.incompleteideas.net/IncIdeas/BitterLesson.html, 2019. Last accessed: 12/03/2025.
- [6] Y. LeCun, Y. Bengio, & G. Hinton, Deep learning, *Nature* **521**:436–444, 2015. doi:10.1038/nature14539
- [7] T. Schmale, M. Reh & M. Gärtner Efficient quantum state tomography with convolutional neural networks, *npj Quantum Inf* **8**:115, 2022. doi:10.1038/s41534-022-00621-4
- [8] G. Torlai et al, Neural-network quantum state tomography, *Nature Physics* **14**:447–450, 2018. doi:10.1038/s41567-018-0048-5
- [9] H. Xiao and Z. Han, Quantum state estimation based on deep learning, *Chin. Phys. B* **33**(12):120307, 2024. doi:10.1088/1674-1056/ad78d7
- [10] S. Lohani et al., On the experimental feasibility of quantum state reconstruction via machine learning, *IEEE* **2**:2103410, 2021. doi:10.1109/TQE.2021.3106958
- [11] S. Wallentowitz and W. Vogel, Reconstruction of the Quantum Mechanical State of a Trapped Ion, *Phys. Rev. Lett.* **75**(16):2932, 1995. doi:10.1103/PhysRevLett.75.2932
- [12] U. Fano, Description of States in Quantum Mechanics by Density Matrix and Operator Techniques, *Rev. Mod. Phys.* **29**:74, 1957. doi:10.1103/RevModPhys.29.74
- [13] Z. Quin et al., Optimal Allocation of Pauli Measurements for Low-Rank Quantum State Tomography, *Trans. Quant. Eng.* **7**:2100416, 2026. doi:10.1109/TQE.2026.3654929
- [14] N. P. Landsman, Born Rule and its Interpretation, *Springer* **1**:64–70, 2009. doi:10.1007/978-3-540-70626-720
- [15] Y-K. Liu, Universal low-rank matrix recovery from pauli measurements, *arXiv:1103.28162011*, 2011, doi: 10.48550/arXiv.1103.2816.
- [16] S. T. Flammia, D. Gross, Y-K. Liu and J. Eisert, Quantum tomography via compressed sensing: error bounds, sample complexity and efficient estimators, *New J. Phys.* **14**:095022, 2012. doi:10.1088/1367-2630/14/9/095022
- [17] K. Georgopolous, C. Emary & P. Zuliani, Modeling and simulating the noisy behavior of near-term quantum computers. *Phys. Rev. A* **104**:062432, 2021. doi:10.1103/PhysRevA.104.062432
- [18] N. Gisin and R. Thew, Quantum communication, *Nature Photon.* **1**:165–171, 2007. doi:10.1038/nphoton.2007.22
- [19] C. Brukner, On the quantum measurement problem, *Springer* pp95–117, doi:10.1007/978-3-319-38987-55
- [20] H-Y. Hsieh, Neural-network-enhanced Fock-state tomography, *Phys. Rev. A* **110**:053705, 2024.

doi:10.1103/PhysRevA.110.053705

- [21] K. Vogel and H. Risken, Determination of quasiprobability distributions in terms of probability distributions for the rotated quadrature phase, *Phys. Rev. A* **40**(5):2847, 1989. doi:10.1103/PhysRevA.40.2847
- [22] D. Leibfried et al., Experimental Determination of the Motional Quantum State of a Trapped Atom, *Phys. Rev. Lett.* **77**(21):4281, 1996. /doi:10.1103/PhysRevLett.77.4281
- [23] J. I. Cirac and P. Zoller, Quantum computations with cold trapped ions, *Phys. Rev. Lett.* **74**:4091, 1995. doi:10.1103/PhysRevLett.74.4091
- [24] Spinquantum, What is entanglement in quantum computing & how it works, <https://www.spinquanta.com/news-detail/entanglement-in-quantum-computing>, 2025. Last accessed 01/04/2026.
- [25] J. S. Bell, Speakable and unspeakable in quantum mechanics, *Cam. Univ. Press* **2**, 2004.
- [26] D. P. DiVincenzo, Two-bit gates are universal for quantum computation, *Phys. Rev. A* **51**:1015, 1995. doi:10.1103/PhysRevA.51.1015
- [27] M. Cramer et al., Efficient quantum state tomography, *Nature Commun.* **1**:149, 2010. doi:10.1038/ncomms1147
- [28] A. L. Samuel, Some studies in machine learning using the game of checkers, *IBM Journal of Research and Development*, **3**(3):210–229, 1959. doi:10.1147/rd.441.0206
- [29] J. Alzubi et al., Machine Learning from Theory to Algorithms: An Overview, *Journal of Physics: Conference Series*, **1142**:012012, 2018. doi:10.1088/1742-6596/1142/1/012012
- [30] T. Phuoc et al., Applying machine learning algorithms to predict the stock price trend in the stock market – The case of Vietnam, *Humanities and Social Sciences Communications*, **11**:393, 2024. doi:10.1057/s41599-024-02807-x
- [31] M. M. Ahsan, S. A. Luna & Z. Siddique, Machine-Learning-Based Disease Diagnosis: A Comprehensive Review. *Healthcare*, **15**(3):541, 2022. doi:10.3390/healthcare10030541
- [32] M. R. Bachute and J. M. Subhedar, Autonomous Driving Architectures: Insights of Machine Learning and Deep Learning Algorithms, *Machine Learning with Applications*, **6**:100164, 2021. doi:10.1016/j.mlwa.2021.100164
- [33] G. S. Hukkeri and R. H. Goudar, Machine Learning-Based Personalized Recommendation System for E-Learners, *IEEE* pp.1–6, 2022. doi:10.1109/ICSTCEE56972.2022.10100069
- [34] Y. LeCun, L. Bottou, Y. Bengio & P. Haffner, Gradient-based learning applied to document recognition, *Proceedings of the IEEE*, **86**(11):2278–2324, 1998. doi:10.1109/5.726791.
- [35] S. R. Dubey, S. K. Singh & B. B. Chaudhuri, Activation functions in deep learning: A comprehensive study and benchmark, *Neurocomputing*, 2022. doi:10.48550/arXiv.2109.14545
- [36] I. D. Mienye, Deep Convolutional Neural Networks: A Comprehensive Review, *Preprints* 2024. doi:10.20944/preprints202408.1288.v1
- [37] P. Feng and Z. Tang, A survey of visual neural networks: current trends, challenges and opportunities, *Multimedia Systems*, **29**(3):693–724, 2023. doi:10.1007/s00530-022-01003-8
- [38] P. J. Forrester and T. Nagao, Eigenvalue Statistics of the Real Ginibre Ensemble, *Phys. Rev. Lett.* **99**:050603, 2007. doi:10.1103/PhysRevLett.99.050603
- [39] K. Mølmer and A. Sørensen, Multiparticle entanglement of hot trapped ions, *Phys. Rev. Lett.* **82**:1835, 1999. doi:10.1103/PhysRevLett.82.1835

- [40] G. Fitzgerald and W. Yeadon, QSTToolkit: A Python Library for Deep Learning Powered Quantum State Tomography, *arXiv preprint* 2503.14422, 2025. doi:10.48550/arXiv.2503.14422
- [41] Z. Hradil et al., Maximum-likelihood methods in quantum mechanics, *Lect. Notes. Phys.* **649**:59–112, 2004. 10.1007/978-3-540-44481-73
- [42] A. Böttcher and D. Wenzel, The Frobenius norm and the commutator, *Elsevier* **429**:8–9, 2008. 10.1016/j.laa.2008.05.020
- [43] T. L. Scholten and R. Blume-Kohout, Behavior of the maximum likelihood in quantum state tomography, *New J. Phys.* **20**: 023050, 2018. doi:10.1088/1367-2630/aaa7e2
- [44] N. Higham, Analysis of the Cholesky Decomposition of a Semi-Definite Matrix, *Oxf. Un. P.*, 1995.
- [45] J. D. Otálora, Bayesian Optimization on Neural Network Hyperparameters, <https://medium.com/bayesian-optimization-on-neural-network-hyperparameters>. Last accessed: 15/03/2026.
- [46] J. Schmidt-Hieber, Nonparametric regression using deep neural networks with ReLU activation function, *Ann. Statist.* **48**(4):1875-1897, 2020. doi:10.1214/19-AOS1875
- [47] S. Weglarczyk, Kernel density estimation and its application, *ITM Web. Conf.* **23**:00037, 2018. doi:10.1051/itmconf/20182300037
- [48] H. Abdi and L. J. Williams, Principal component analysis, *WIREs Comp. Stats.* **2**(4), 2010. doi:10.1002/wics.101
- [49] G. Kurnia, Understanding cosine similarity: a powerful tool for measuring vector similarity, 2024. <https://medium.com/@gunkurnia/understanding-cosine-similarity-a-powerful-tool-for-measuring-vector-similarity>. Last accessed: 12/02/2026.
- [50] F. Schmidt-Kaler, Realization of the Cirac-Zoller controlled-NOT quantum gate, *Nature* **422**:408–411, 2003. doi:10.1038/nature01494
- [51] H. Häffner et al., Scalable multiparticle entanglement of trapped ions, *Nature* **438**:04279, 2005. doi:10.1038/nature04279
- [52] Q. A. Turchette et al., Deterministic entanglement of two trapped ions, *Phys. Rev. Lett.* **81**(17):3631–3634, 1998. doi:10.1103/PhysRevLett.81.3631
- [53] G. C. Knee et al., Maximum-likelihood quantum process tomography via projected gradient descent, *arXiv preprint* 1803.10062, 2018. doi:arXiv:1803.10062
- [54] D. Gneebaum, Introduction to Quantum Gate Set Tomography, *arXiv preprint* 1509.02921, 2015. doi:arXiv:1509.02921v1
- [55] J. Rehacek and Z. Hradil, Diluted maximum-likelihood algorithm for quantum tomography, *Phys. Rev. A* **75**:042108, 2007. doi:10.1103/PhysRevA.75.042108
- [56] L. S. Madsen, Quantum computational advantage with a programmable photonic processor, *Nature* **606**:75–81, 2022. doi:10.1038/s41586-022-04725-x
- [57] J. G. Titchener, Scalable on-chip quantum state tomography, *npj Quant. Inf.* **4**:19, 2018. doi:10.1038/s41534-018-0063-5
- [58] Quantinuum: Quantinuum Launches Industry-First, Trapped-Ion 56-Qubit Quantum Computer, Breaking Key Benchmark Record, 2024. <https://www.quantinuum.com/press-releases/>. Last accessed: 19/04/2026

SCIENTIFIC SUMMARY FOR A GENERAL AUDIENCE

Quantum physics describes nature at the smallest (sub-atomic) scales, where systems exist in multiple states simultaneously until measured. Because a single measurement “collapses” this state and destroys the remaining information, scientists use Quantum State Tomography (QST) - repeatedly measuring many identical copies of the system - to reconstruct its full picture.

In particular, it is often necessary to check whether an experiment has produced the intended state. However, QST faces two hurdles: environmental “noise” that distorts the measured state, and a steep increase in computational cost as systems grow larger. This study investigated whether Deep Neural Networks (NNs), a form of artificial intelligence, could overcome these challenges by learning to predict the full quantum state from imperfect measurement data.

When compared against traditional QST methods, the NN was significantly faster and more accurate across all tested conditions, showcasing its ability to learn patterns that helped it separate genuine features from noise. Crucially, the NN successfully predicted states it hadn’t encountered during training, indicating it had learned to solve the underlying task rather than simply memorising examples.

Ultimately, this research demonstrates that AI is a powerful tool for measuring complex quantum systems. This follows the general trend in science of AI-driven computation tackling problems that are difficult to achieve with only human-led methods.

STATEMENT ON USE OF GENERATIVE AI

Generative AI tools were used on occasion for syntax and debugging of the computational program, and to check phrasing, spelling, and grammar to improve the overall clarity of the report. Their inclusion did not extend to the technical content nor any original research ideas.

Appendix A: Additional Quantum Mechanics Background

A quantum system is described mathematically by a state, which contains all information that can be known about the system in a chosen measurement basis. For a pure state this is represented by $|\psi\rangle \in \mathcal{H}$, where $\mathcal{H}$ is the Hilbert space of the system. More generally, especially in tomography, states are represented by a density operator ρ , which is Hermitian, positive semidefinite, and satisfies $\text{Tr}(\rho) = 1$. For an n -qubit system, the Hilbert-space dimension is $d = 2^n$.

A single quantum measurement is not sufficient to determine the full state. A projective measurement yields only one outcome and, in general, disturbs the state by projecting it onto an eigenspace of the measured observable. Full state estimation therefore requires many independently prepared copies of the same state, measured in different bases.

Two important principles underlying this are:

- **Heisenberg uncertainty principle:** pairs of non-commuting observables, such as position and momentum, cannot be known simultaneously with arbitrary precision.
- **No-cloning theorem:** an unknown quantum state cannot be copied perfectly, so repeated measurements must be performed on many separately prepared identical states rather than on a single state.

The basic unit of quantum information is the qubit, a two-level quantum system with computational basis states $|0\rangle$ and $|1\rangle$. A general pure qubit state may be written

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1, \quad (\text{A1})$$

and can be visualised geometrically on the Bloch sphere.

Quantum measurements are described by positive operator-valued measures (POVMs). A POVM is a set of positive semidefinite operators $\{E_m\}$ satisfying

$$E_m \geq 0, \quad \sum_m E_m = I, \quad (\text{A2})$$

with outcome probabilities given by the Born rule

$$p(m) = \text{Tr}(\rho E_m). \quad (\text{A3})$$

In this work, tomography is performed using the two-outcome POVM associated with each non-identity n -qubit Pauli operator P_k :

$$E_{k,\pm} = \frac{1}{2} (I \pm P_k), \quad (\text{A4})$$

corresponding to projection onto the ± 1 eigenspaces of P_k . For the two-qubit case there are $4^2 - 1 = 15$ non-identity Pauli observables, giving 30 POVM elements in total.

Figure 9 shows the full two-qubit POVM set used in the tomography simulations.

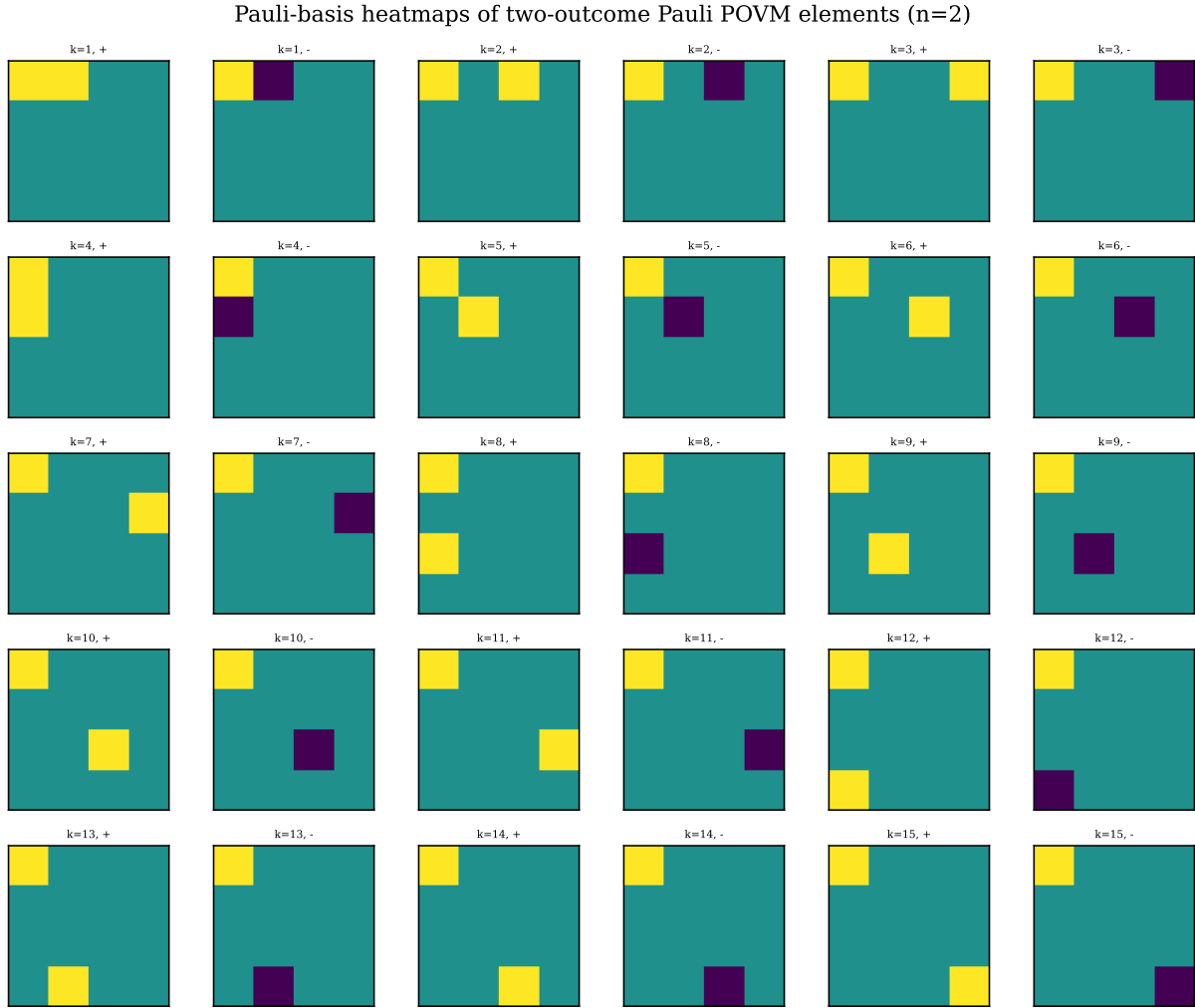


FIG. 9: Graphical representation of the two-qubit Pauli POVM elements used for tomography. The figure is arranged as 30 heatmaps, one for each two-outcome POVM element $E_{k,\pm}$, grouped by the 15 non-identity two-qubit Pauli observables P_k . Each individual 4×4 heatmap represents the corresponding matrix operator in the two-qubit computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$. The colour scale shows the matrix entries: green denotes 0, yellow denotes +1, and purple denotes -1.

Appendix B: Deep Learning Neural Networks

1. Additional Deep Learning Background

A neural network is a computational model that maps an input vector to an output vector through a sequence of layers of artificial neurons. Each layer applies an affine transformation followed by a nonlinear activation function, allowing the network to approximate complex non-linear mappings. In supervised learning, the parameters of the network are adjusted to minimise a loss function measuring the discrepancy between predicted and target outputs.

Several activation functions are commonly used in modern neural networks. The sigmoid and hyperbolic tangent ($\tanh$) functions were historically important, but can suffer from van-

ishing gradients in deep architectures. In contrast, the rectified linear unit (ReLU),

$$\text{ReLU}(x) = \max(0, x), \quad (\text{B1})$$

is now widely used because of its computational simplicity and strong empirical performance. For regression problems, such as the present prediction of density-matrix elements, the output layer is typically taken to be linear.

The optimisation of network parameters is carried out using gradient-based algorithms. Standard stochastic gradient descent updates the model parameters using mini-batches of training data. More advanced optimisers, such as Adam, adapt the learning step for each parameter individually using running estimates of first and second moments of the gradient. Adam is widely used in practice because it is robust and performs well across a wide range of learning tasks.

To improve generalisation, neural networks often employ regularisation techniques. Two common approaches are L_2 regularisation and dropout. L_2 regularisation penalises large weight magnitudes during training, encouraging smoother solutions, while dropout randomly deactivates a fraction of neurons during training to reduce overfitting and discourage the network from relying too strongly on any one subset of learned features.

2. Methodology: Hyperparameter Tuning

Several candidate MLP architectures were tested in order to establish a suitable baseline architecture for the tomography task. These models differed in layer width and regularisation strategy, while preserving the same input and output structure. All models were trained on the representative noisy two-qubit dataset introduced in the main text, and their mean fidelity was used as the primary comparison metric. The five candidate models are summarised in Table III.

TABLE III: Candidate neural-network models considered during architecture selection.

Model	Layers	L_2 regularisation	Dropout	Learning rate
1	512, 128	0	0	1.0×10^{-3}
2	256, 256, 128	0.001	0	1.0×10^{-3}
3	256, 256, 128	0	0	1.0×10^{-3}
4	1024, 1024	0	0	1.0×10^{-3}
5	256, 256, 128	0	0.2	1.5567×10^{-3}

Here, L_2 regularisation adds a penalty proportional to the squared weight norm to the loss function, discouraging excessively large parameter values. Dropout gives the fraction of hidden neurons randomly deactivated during each training step. The learning rate controls the size of each parameter update during optimisation.

Figure 10 compares the performance of the candidate models. The large reference model performs substantially worse and with much higher variability than the smaller three-layer architectures, indicating that simply increasing parameter count does not improve the task. The best-performing models are those based on the 256-256-128 architecture, with dropout giving the highest mean fidelity and strongest stability. This motivated the choice of the dropout model for the main investigation.

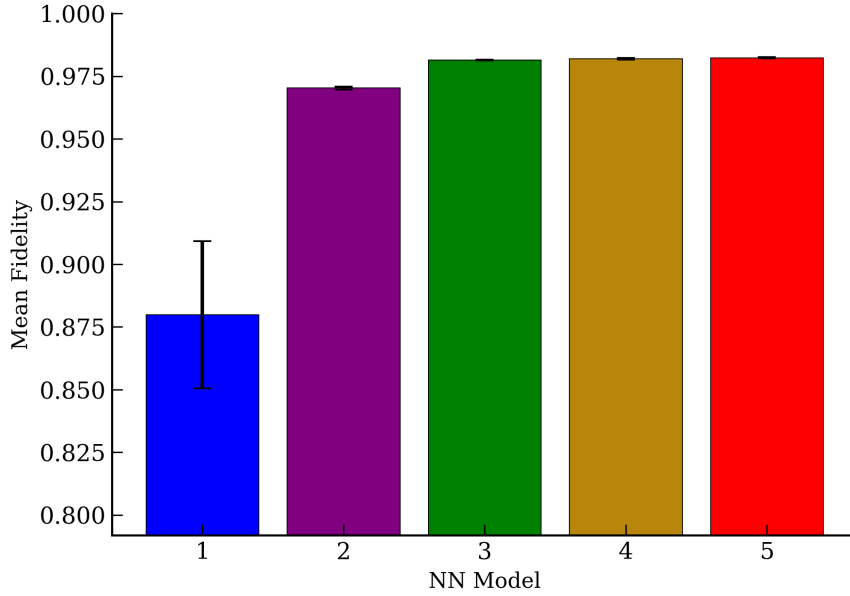


FIG. 10: Comparison of the five candidate neural-network models. Bars show mean fidelity on the representative noisy two-qubit benchmark, with error bars denoting one standard deviation over repeated runs.

L2 regularisation didn't improve performance over the standard model (3), perhaps by penalising useful weight information. The reference model with the 1024-1024 layers [9] demonstrated competitive fidelity, but demonstrated significantly greater overfitting than the three-layered architecture, shown in Figure 11. Since it did also not outperform model 3 significantly, it was accordingly discarded.

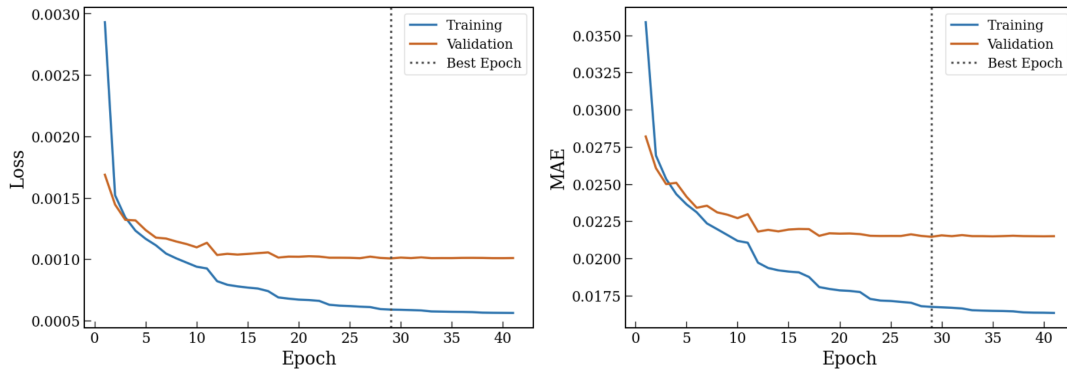


FIG. 11: MSE and MAE training and validation loss curves for the reference model (4). Significant overfitting is evident.

Having fixed the general architecture, Bayesian optimisation was used to refine the learning rate and regularisation strength. Bayesian optimisation tunes hyperparameters by fitting a probabilistic surrogate model to the results of previous trials and then using that model to predict which region of parameter space is most likely to improve performance. Unlike grid search, which evaluates a fixed set of points independently, Bayesian optimisation uses information from earlier trials to choose the next one more intelligently, making it much more efficient

when training runs are expensive. This optimisation was restricted to these two hyperparameters because they provide the most direct control over training stability and generalisation while keeping the search computationally tractable and stable. Extending Bayesian optimisation to all architectural choices, such as layer widths and depth, would greatly enlarge the search space and make the procedure noisy and expensive for comparatively limited gain at this stage. The optimisation therefore acted as a refinement of an already suitable architecture rather than as a complete architecture search.

The final model selected for the main study used the 256-256-128 architecture with dropout regularisation (0.2) and a Bayesian-optimised learning rate of 1.5567×10^{-3} - Model 5.

Finally, the size of the training dataset was chosen by monitoring validation-loss convergence as the number of training states increased. Figure 12 shows that the validation loss decreases rapidly as the training set is enlarged, before gradually approaching a plateau at larger dataset sizes. The corresponding best epoch is also shown. This indicates that the chosen dataset size is sufficiently large to stabilise training performance without incurring unnecessary computational cost.

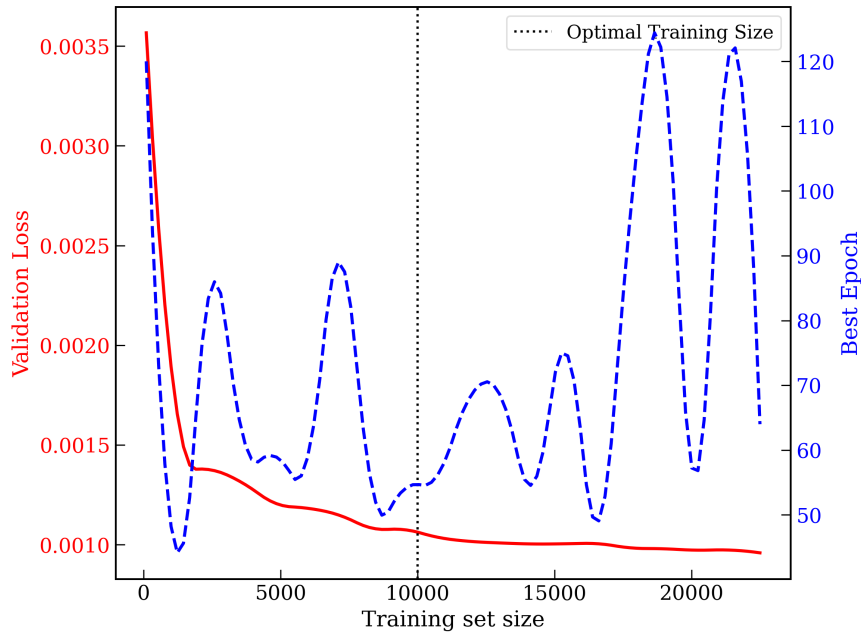


FIG. 12: Convergence of validation loss with training-set size for the selected architecture. The red curve shows the minimum validation loss obtained for each dataset size, while the blue dashed curve shows the corresponding best epoch.

Listing 1: Computational impenetation of the dropout model (5) in Tensorflow.Keras

```
def build_mlp_drop(n_qubits):
    input_dim = 4**n_qubits
    output_dim = 2 * (2**n_qubits)**2

    inputs = keras.Input(shape=(input_dim,))
    x = layers.Dense(256, activation="relu")(inputs)
```

```

x = layers.Dropout(0.2)(x)
x = layers.Dense(256, activation="relu")(x)
x = layers.Dropout(0.2)(x)
x = layers.Dense(128, activation="relu")(x)
outputs = layers.Dense(output_dim, activation="linear")(x)
model = keras.Model(inputs, outputs, name=f"mlp_qst_{n_qubits}qubit")
model.compile(
    optimizer=keras.optimizers.Adam(learning_rate=0.0015567),
    loss="mse",
    metrics=["mae"]
)
return model

```

3. Additional Neural Network Interpretability

A secondary view of the cosine-similarity analysis is obtained by examining which physical projector is selected most often as the best match for each hidden neuron. For every neuron in a given layer, the two-qubit Pauli projector with the largest absolute cosine similarity was identified, and the frequencies of these best-matched projector labels were then counted across the full layer. Figure 13 shows the resulting distributions for the three dense hidden layers of the final dropout model.

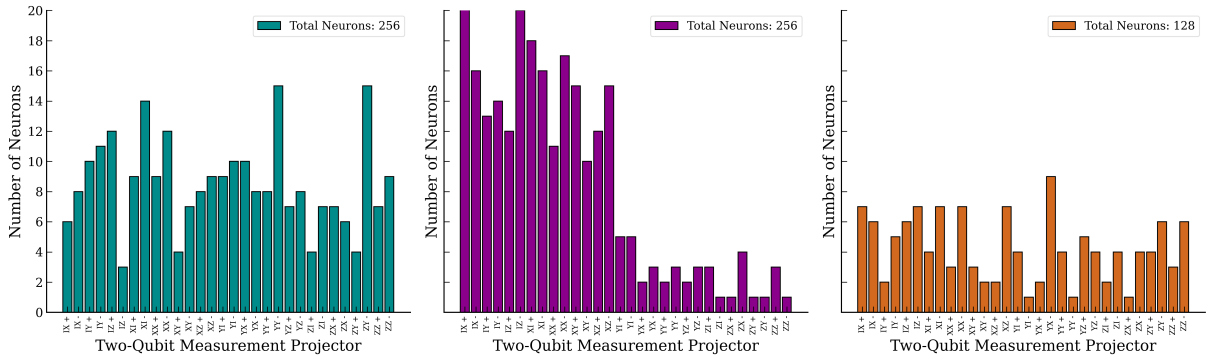


FIG. 13: Frequency distributions of the best-matched two-qubit Pauli-projector labels for neurons in the three dense hidden layers of the final dropout model. For each neuron, the projector with the largest absolute cosine similarity was identified. The three panels correspond to the first, second, and third hidden layers respectively.

The second hidden layer shows the clearest concentration on a more restricted subset of projectors. This indicates that some intermediate compression or feature selection is taking place within the network, with certain measurement directions becoming more prominent than others. However, this should not be interpreted as recovery of a true Pauli-projector basis. The corresponding cosine-similarity values remain only moderate, so the concentration seen here

is better understood as evidence that some learned features are more strongly associated with particular projectors, rather than that the layer itself has become physically interpretable in tomographic terms.

The first and third hidden layers shows a broader spread. The absence of a sharp peak on a single projector family suggests that, even close to the output stage, the network still relies on a distributed representation rather than on a one-to-one correspondence with the physical measurement operators.

Taken together, these frequency plots support the same conclusion as the cosine-similarity distributions: the hidden layers do not reconstruct a simple projector-aligned tomography basis. Instead, the network develops partial sensitivity to certain measurement directions while retaining a broadly distributed internal representation. This is consistent with its role as a supervised denoising estimator, where the objective is not to reproduce the structure of Stokes reconstruction or MLE, but to learn whichever internal features best map noisy measurements to accurate density-matrix predictions.

Appendix C: Dataset Generation and State Families

This appendix summarises the computational implementation used to generate the synthetic quantum state datasets employed throughout this work. The framework is designed to produce valid quantum states from several state families depending on the physical system or experiment being modelled, including circuit-generated pure states, Haar-random pure states, Ginibre mixed states, and selected named states such as Bell, GHZ, and W states.

1. Single-Qubit Operators and Tensor Products

The single-qubit identity and Pauli operators are defined as

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (\text{C1})$$

For n -qubit systems, multi-qubit operators are constructed by tensor products of these single-qubit operators. If $M_1, \dots, M_n$ act on the individual qubits, then

$$P_1 \otimes P_2 \otimes \dots \otimes P_n \quad (\text{C2})$$

acts on the full Hilbert space $\mathcal{H}_{2^n}$.

The action of a single-qubit operator on qubit q within an n -qubit register is implemented as:

Listing 2: Embedding a single-qubit operator into an n -qubit register.

```
I = np.array([[1,0],[0,1]], complex)
X = np.array([[0,1],[1,0]], complex)
Y = np.array([[0,-1j],[1j,0]], complex)
Z = np.array([[1,0],[0,-1]], complex)
PAULI_1Q = [I, X, Y, Z]
```

```

def kron_all(mats):
    out = mats[0]
    for M in mats[1:]:
        out = np.kron(out, M)
    return out
def op_on_qubit(single_op, n, q):
    return kron_all([single_op if i == q else I for i in range(n)])

```

2. Density-Matrix Construction

Given a pure state vector $|\psi\rangle$, the corresponding density matrix is

$$\rho = |\psi\rangle \langle \psi|. \quad (\text{C3})$$

For numerical use, density matrices are also vectorised by concatenating their real and imaginary parts:

$$\rho \mapsto \mathbf{v}_\rho \in \mathbb{R}^{2d^2}, \quad d = 2^n. \quad (\text{C4})$$

This representation is used as the supervised-learning target for the neural network.

3. State Families

Haar-Random Pure States

A Haar-random pure state is generated by sampling a complex vector

$$|\psi\rangle = (z_1, \dots, z_d)^T, \quad z_j \sim \mathcal{N}(0, 1) + i\mathcal{N}(0, 1), \quad (\text{C5})$$

and normalising it:

$$|\psi\rangle \leftarrow \frac{|\psi\rangle}{\| |\psi\rangle \|}. \quad (\text{C6})$$

The density matrix is then $\rho = |\psi\rangle \langle \psi|$.

The corresponding Python function is:

Listing 3: Generation of Haar-random pure states.

```

def haar_pure_rho(n, rng):
    d = 2*n
    psi = rng.normal(size=d) + 1j*rng.normal(size=d)
    psi /= np.linalg.norm(psi)
    return rho_from_ket(psi)

```

Ginibre Mixed States

A random mixed state is generated from the Ginibre ensemble by first sampling a complex matrix

$$G \in \mathbb{C}^{d \times k}, \quad G_{ij} \sim \mathcal{N}(0, 1) + i\mathcal{N}(0, 1), \quad (\text{C7})$$

then forming

$$A = GG^\dagger, \quad \rho = \frac{A}{\text{Tr}(A)}. \quad (\text{C8})$$

When $k = d$, this gives a generic random mixed state.

This is implemented as:

Listing 4: Generation of Ginibre mixed states.

```
def ginibre_mixed_rho(n, rng, k=None):
    d = 2**n
    if k is None:
        k = d
    G = rng.normal(size=(d,k)) + 1j*rng.normal(size=(d,k))
    A = G @ G.conj().T
    return A / np.trace(A)
```

Bell States

The maximally-entangled two-qubit Bell states are defined

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle), \quad (\text{C9})$$

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}} (|00\rangle - |11\rangle), \quad (\text{C10})$$

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}} (|01\rangle + |10\rangle), \quad (\text{C11})$$

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}} (|01\rangle - |10\rangle). \quad (\text{C12})$$

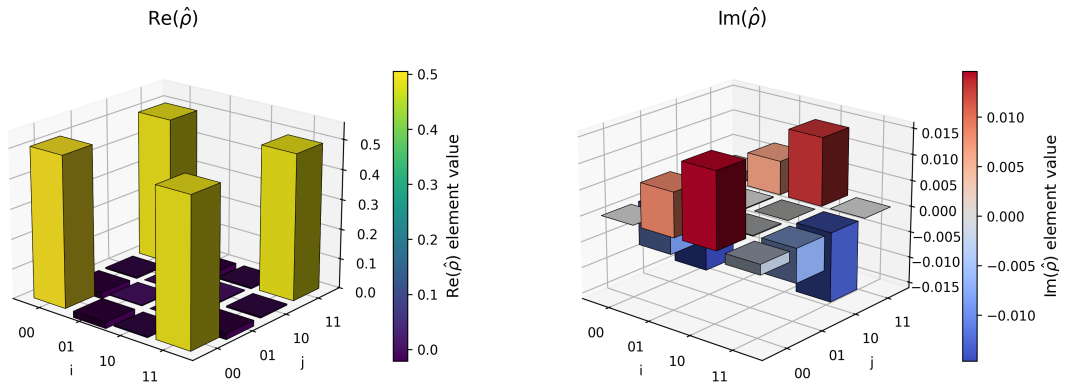


FIG. 14: NN-reconstructed $|\Phi^+\rangle$ Bell state under the trapped ion-inspired error budget ($F = 0.99$).

GHZ States

The n -qubit GHZ state is defined as

$$|\text{GHZ}_n\rangle = \frac{1}{\sqrt{2}} (|0\rangle^{\otimes n} + |1\rangle^{\otimes n}). \quad (\text{C13})$$

W States

The n -qubit W state is defined as

$$|W_n\rangle = \frac{1}{\sqrt{n}} \sum_{k=1}^n |0\rangle^{\otimes(k-1)} |1\rangle |0\rangle^{\otimes(n-k)}. \quad (\text{C14})$$

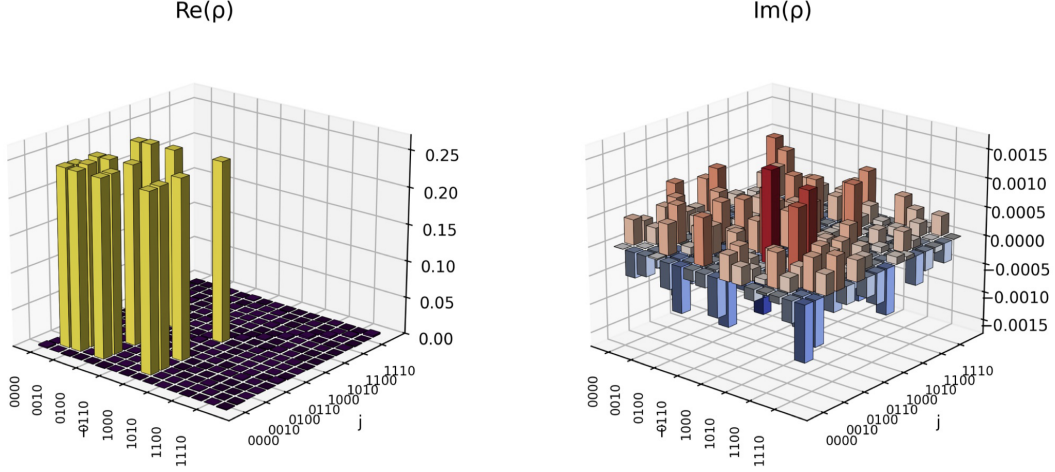


FIG. 15: NN-reconstructed 4-qubit W state under the trapped ion-inspired error budget ($F = 0.96$).

4. Trapped Ion-Inspired Circuit-Generated Pure States

The main synthetic dataset used in this work is generated by a randomised circuit inspired by trapped-ion hardware. The initial register is prepared in the computational ground state,

$$|\psi_0\rangle = |0\rangle^{\otimes n}. \quad (\text{C15})$$

Single-Qubit Rotations

Single-qubit rotations about the X , Y , and Z axes are defined as

$$R_X(\theta) = e^{-i\theta X/2}, \quad R_Y(\theta) = e^{-i\theta Y/2}, \quad R_Z(\theta) = e^{-i\theta Z/2}. \quad (\text{C16})$$

When applied globally to all n qubits, these become

$$U_X(\theta) = R_X(\theta)^{\otimes n}, \quad U_Y(\theta) = R_Y(\theta)^{\otimes n}, \quad U_Z(\theta) = R_Z(\theta)^{\otimes n}. \quad (\text{C17})$$

The global single-qubit rotations used in the circuit are implemented as:

Listing 5: Global basis rotations applied uniformly to all qubits.

```
def global_rotation(n, axis, theta):
    '''Rotates all qubits together'''
    if axis == "X":
        s = X
```

```

elif axis == "Y":
    s = Y
elif axis == "Z":
    s = Z
else:
    raise ValueError("axis must be X/Y/Z")
U1 = scipy.linalg.expm(-1j * theta/2 * s)
return kron_all([U1 for _ in range(n)])

```

All-to-All XX -Type Entangler

The effective entangling gate used in the circuit is an all-to-all XX -type unitary of the form

$$U_{XX}(\theta) = \exp\left(-\frac{i\theta}{2} \sum_{i<j} X_i X_j\right), \quad (\text{C18})$$

where X_i denotes the Pauli- X operator acting on qubit i . For two qubits, this reduces to the familiar R_{XX} gate,

$$R_{XX}(\theta) = \exp\left(-\frac{i\theta}{2} X \otimes X\right). \quad (\text{C19})$$

Listing 6: Effective all-to-all XX -type entangling gate.

```

def ms_entangler_effective(n, theta):
    H = np.zeros((2**n, 2**n), complex)
    for i in range(n):
        Xi = op_on_qubit(X, n, i)
        for j in range(i+1, n):
            Xj = op_on_qubit(X, n, j)
            H += Xi @ Xj
    return scipy.linalg.expm(-1j * (theta/2) * H)

```

Circuit Structure

A depth- D random circuit is built from alternating global single-qubit rotations and the XX -type entangler. The total unitary is

$$U_\psi = \prod_{\ell=1}^D \left[U_{XX}(\theta_{XX}^{(\ell)}) \left(\bigotimes_{i=1}^n R_X(\theta_X^{(\ell)}) \right) \left(\bigotimes_{i=1}^n R_Y(\theta_Y^{(\ell)}) \right) \right]. \quad (\text{C20})$$

The circuit depth is sampled from

$$D \sim \mathcal{U}\{1, 2, 3\}, \quad (\text{C21})$$

and the gate parameters are drawn independently at each layer according to

$$\theta_X^{(\ell)} \sim \mathcal{U}(0, 2\pi), \quad \theta_Y^{(\ell)} \sim \mathcal{U}(0, 2\pi), \quad \theta_{XX}^{(\ell)} \sim \mathcal{U}(0, \pi/2). \quad (\text{C22})$$

The final state is therefore

$$|\psi\rangle = U_\psi |\psi_0\rangle, \quad \rho = |\psi\rangle \langle\psi|. \quad (\text{C23})$$

This construction guarantees that all generated circuit states are valid pure states, since the initial state is normalised and all gates are unitary.

The computational implementation of the unitary circuit is:

Listing 7: Ion-inspired circuit state generation.

```
def ion_like_circuit_rho(n, rng, depth=2):
    d = 2**n
    psi = np.zeros(d, complex)
    psi[0] = 1.0

    U = np.eye(d, dtype=complex)
    for _ in range(depth):
        U = global_rotation(n, "Y", rng.uniform(0, 2*np.pi)) @ U
        U = global_rotation(n, "X", rng.uniform(0, 2*np.pi)) @ U
        U = ms_entangler_effective(n, rng.uniform(0, np.pi/2)) @ U

    psi = U @ psi
    return rho_from_ket(psi)
```

5. State-Family Sampling Procedure

The dataset generator includes a sampler that selects among several state families according to user-defined mixture weights. In the implementation used here, the selected family may be:

- a named state family (Bell, GHZ, or W),
- an ion-inspired circuit state,
- a Haar-random pure state,
- or a Ginibre mixed state.

For a given qubit number n , the dataset is then formed by repeatedly sampling states from the chosen family mixture and storing:

1. the Pauli expectation vector,
2. and the vectorised target density matrix.

Although the ion-inspired circuit family provides the main state ensemble used in this work, the inclusion of multiple state-generation procedures ensures that the computational framework

is not restricted to one specific preparation model. This flexibility supports the broader methodological aim of developing and benchmarking QST methods that can, in principle, be extended to a range of physical platforms and state classes.

Each state class can be generated with a probability determined by the user. The framework includes a sampler that selects between named states, ion-inspired circuit states, Haar-random pure states, and Ginibre mixed states according to user-defined mixture weights:

Listing 8: State-family sampler used in dataset generation.

```
def sample_trapped_ion_state(n, rng, mix=None):
    if mix is None:
        mix = {"named": 0.0, "ion": 1.0, "haar": 0.0, "ginibre": 0.0}

    keys = list(mix.keys())
    w = np.array([mix[k] for k in keys], float)
    w = w / w.sum()
    choice = rng.choice(keys, p=w)

    if choice == "named":
        if n == 2:
            psi = ket_bell(rng.choice(["phi+", "phi-", "psi+", "psi-"]))
            return rho_from_ket(psi)
        fam = rng.choice(["ghz", "w"])
        psi = ket_ghz(n) if fam == "ghz" else ket_w(n)
        return rho_from_ket(psi)

    if choice == "ion":
        return ion_like_circuit_rho(n, rng, depth=rng.integers(1, 4))

    if choice == "haar":
        return haar_pure_rho(n, rng)

    if choice == "ginibre":
        return ginibre_mixed_rho(n, rng)

    raise ValueError("Bad mix choice.")
```

The final dataset is generated by repeatedly sampling states from the chosen family mixture, computing their Pauli expectation values, and storing the corresponding vectorised density matrices as targets.

Listing 9: Construction of the n -qubit dataset.

```
def make_dataset_nq(N, n_qubits, seed=0, mix=None):
    rng = np.random.default_rng(seed)
    d = 2**n_qubits
```

```

X = np.zeros((N, 4*n_qubits), dtype=float)
y = np.zeros((N, 2*(d*d)), dtype=float)

for i in range(N):
    rho = sample_trapped_ion_state(n_qubits, rng, mix=mix)
    X[i] = pauli_expectations_nq(rho, n_qubits)
    y[i] = rho_to_real_vector_nq(rho)
return X, y

```

Appendix D: Trapped-Ion Processor: Physical Implementation

This appendix summarises the $^{40}\text{Ca}^+$ trapped-ion platform used as the physical reference for the scalability benchmark, together with the effective error budget and its mapping onto the simulation channels employed in this work. The purpose here is not to construct a fully device-specific microscopic model, but rather to ground the final scalability benchmark in a realistic experimental setting while preserving the broader methodological scope of the study.

1. Qubit Encoding, Control, and Readout

In the $^{40}\text{Ca}^+$ trapped-ion platform, qubits are encoded in two long-lived internal electronic states of an individual ion. In this work, the computational basis is identified as

$$|0\rangle \equiv |S_{1/2}\rangle, \quad |1\rangle \equiv |D_{5/2}\rangle, \quad (\text{D1})$$

where $|S_{1/2}\rangle$ is the electronic ground state and $|D_{5/2}\rangle$ is a metastable excited state. State initialisation is performed by optical pumping into the ground state, after which coherent laser pulses near 729 nm drive single-qubit rotations on the quadrupole transition between these levels. Multi-qubit entangling operations are mediated through the shared motional modes of the ion chain, for example through Mølmer-Sørensen interactions. Readout is performed by state-dependent fluorescence: laser light resonant with the $S_{1/2} \leftrightarrow P_{1/2}$ transition causes ions in the bright state to scatter photons, while ions in the dark $D_{5/2}$ state remain non-fluorescent.

2. Derived Error Budget

The final scalability benchmark refines the earlier phenomenological noise models into a trapped-ion-inspired operating point. Rather than reproducing the full experimental apparatus in microscopic detail, the dominant reported error sources are mapped onto a small number of effective simulation channels that capture their qualitative action at the preparation, control, measurement, and readout stages.

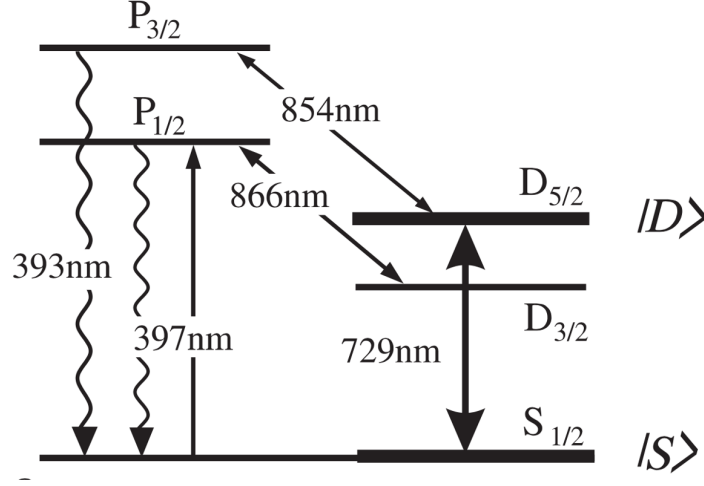


FIG. 16: Electronic structure of the $^{40}\text{Ca}^+$ trapped-ion qubit. The computational basis states are encoded in the $S_{1/2}$ and $D_{5/2}$ levels, coherent control is performed on the 729 nm transition, and state-dependent fluorescence readout is obtained via excitation of the $S_{1/2} \leftrightarrow P_{1/2}$ dipole transition near 397 nm .

TABLE IV: Relevant entries from the published error budget of the $^{40}\text{Ca}^+$ trapped-ion processor used to motivate the scalability benchmark [50]. Only the dominant preparation and control errors relevant to the present simulation are shown.

Error source	Magnitude	Contribution
Laser frequency noise (phase coherence)	$\sim 100\text{ Hz (FWHM)}$	10%
Laser intensity fluctuations	$\sim 3\%$ peak-to-peak	$\sim 1\%$
Laser detuning error	$\sim 200\text{ Hz}$	$\sim 2\%$
Residual thermal excitation	$\langle n \rangle_{\text{other}} < 6$	0.4%
Addressing error	5% in Rabi frequency (neighbouring ion)	3%
Finite measurement cycles	100 shots	-
Fluorescence missclassification	0.02	-

3. Channel Definitions

Coherent Overrotation

In the abstract two-qubit sweeps, preparation and control imperfections were represented phenomenologically by additive Gaussian perturbations to the Pauli expectation values. For the trapped-ion-inspired scalability benchmark, this abstract perturbation is refined into an explicit state-level channel representing imperfect coherent control. In this model, each qubit undergoes a small random unitary misrotation about a chosen axis $A \in \{X, Y, Z\}$:

$$U_{\text{rot}} = \bigotimes_{i=1}^n \exp\left(-\frac{i\delta\theta_i}{2}\sigma_A^{(i)}\right), \quad \delta\theta_i \sim \mathcal{N}(0, \sigma_\theta^2), \quad (\text{D2})$$

so that the state transforms as

$$\rho \mapsto \mathcal{E}_{\text{rot}}(\rho) = U_{\text{rot}} \rho U_{\text{rot}}^\dagger. \quad (\text{D3})$$

The parameter σ_θ therefore sets the width of the random misrotation distribution and quantifies the strength of coherent control error. In the present work, $\sigma_\theta = 0.08$ is used in the scalability benchmark.

Local Phase-Flip Channel

To capture dephasing-type control errors such as laser frequency noise and detuning instability, a local phase-flip channel is applied independently to each qubit. For a single qubit, this channel is

$$\mathcal{E}_\phi(\rho) = (1 - p_\phi) \rho + p_\phi Z \rho Z, \quad (\text{D4})$$

with Kraus operators

$$K_0 = \sqrt{1 - p_\phi} I, \quad K_1 = \sqrt{p_\phi} Z. \quad (\text{D5})$$

Applied independently across an n -qubit register, this channel leaves computational-basis populations unchanged while suppressing off-diagonal coherence. The parameter p_ϕ is therefore interpreted as the probability of a phase-flip event on each qubit, and is set to $p_\phi = 0.06$ in the present benchmark.

Finite-Shot Projection Noise

Projection noise is retained unchanged from the general methodology, since it arises directly from the repeated measurement process itself. For each measured Pauli observable P_k , the probability of obtaining the $+1$ outcome is

$$p_{k,+} = \frac{1 + \langle P_k \rangle}{2}, \quad (\text{D6})$$

and the corresponding counts are sampled from

$$n_{k,+} \sim \text{Bin}(N_{\text{shots}}, p_{k,+}), \quad n_{k,-} = N_{\text{shots}} - n_{k,+}. \quad (\text{D7})$$

For the trapped-ion-inspired operating point, the shot number is fixed at

$$N_{\text{shots}} = 100, \quad (\text{D8})$$

consistent with repeated state-detection cycles in the reference experiment.

Bit-Flip Readout Error

In the earlier abstract setting, readout corruption was modelled using pepper noise, in which a fraction of observables was replaced by uninformative counts. For the trapped-ion-inspired benchmark, this is refined into a bit-flip readout channel that more directly represents fluorescence misclassification and cross-talk. If a measurement produces counts n_+ and n_- , then a fraction of the detected outcomes is flipped between the bright and dark channels:

$$k_+ \sim \text{Bin}(n_+, p_{\text{readout}}), \quad k_- \sim \text{Bin}(n_-, p_{\text{readout}}), \quad (\text{D9})$$

and the corrupted counts become

$$n'_+ = n_+ - k_+ + k_-, \quad n'_- = n_- - k_- + k_+. \quad (\text{D10})$$

Thus, f_{flip} represents the probability that an individual measurement outcome is misassigned at readout. In the present benchmark, this is fixed at

$$f_{\text{flip}} = 0.02. \quad (\text{D11})$$

The trapped-ion-inspired benchmark should be understood as a refinement of the earlier abstract noise framework rather than a separate reconstruction methodology. The additive Gaussian preparation/control perturbation is replaced by the more physically interpretable coherent overrotation and phase-flip channels, while pepper readout noise is replaced by a bit-flip readout process that more closely reflects fluorescence misclassification. Finite-shot sampling is retained unchanged. This provides a grounded operating point for the scalability analysis without restricting the broader methodological conclusions of the study to one specific hardware platform.

Appendix E: Extended Results & Discussion

1. Extended Two-Qubit Noise-Sweep Results

Full numerical results underlying the two-qubit noise-sweep analysis of Section 4 B are presented in Table VI.

To summarise the relative advantage of the neural network across each full noise sweep, the normalised fidelity improvement of the NN over a comparison method $X \in \{\text{Stokes}, \text{MLE}\}$ was defined pointwise as

$$I_i = 100 \frac{\bar{F}_{\text{NN},i} - \bar{F}_{X,i}}{\bar{F}_{X,i}}, \quad (\text{E1})$$

where $\bar{F}_{\text{NN},i}$ and $\bar{F}_{X,i}$ are the mean fidelities at the i -th noise value. The normalised improvement reported in Table V is then the arithmetic mean of these pointwise improvements over the full sweep,

$$\bar{I} = \frac{1}{N} \sum_{i=1}^N I_i^{(\text{NN},X)}, \quad (\text{E2})$$

where $N = 6$ is the number of noise points in each sweep.

The uncertainty on each pointwise improvement was obtained by standard first-order error propagation,

$$\sigma_{I_i} = 100 \sqrt{\left(\frac{\sigma_{\text{NN},i}}{\bar{F}_{X,i}} \right)^2 + \left(\frac{\bar{F}_{\text{NN},i} \sigma_{X,i}}{\bar{F}_{X,i}^2} \right)^2}, \quad (\text{E3})$$

where $\sigma_{\text{NN},i}$ and $\sigma_{X,i}$ are the corresponding standard deviations. Assuming independence between noise points, the uncertainty on the mean normalised improvement was estimated as

$$\sigma_{\bar{I}} = \frac{1}{N} \sqrt{\sum_{i=1}^N \sigma_{I_i}^2}. \quad (\text{E4})$$

TABLE V: Normalised fidelity improvement of the NN over MLE and Stokes reconstruction, averaged across the full range of each noise process.

Noise process	NN vs MLE (%)	NN vs Stokes (%)
Additive Gaussian (σ)	5.87 ± 0.06	6.52 ± 0.06
Finite projection (N_{shots})	2.98 ± 0.04	3.51 ± 0.03
Pepper readout (p)	13.01 ± 0.11	8.56 ± 0.11

TABLE VI: Full numerical fidelity results for the two-qubit noise sweeps. Reported uncertainties are standard deviations over repeated runs.

Noise process	Parameter	Stokes		MLE		NN	
		$\bar{F}$	σ_F	$\bar{F}$	σ_F	$\bar{F}$	σ_F
Additive Gaussian	$\sigma = 0.00$	0.999775	2.68×10^{-7}	0.991927	1.47×10^{-5}	0.998596	1.24×10^{-4}
	$\sigma = 0.06$	0.989906	3.83×10^{-5}	0.972703	2.21×10^{-4}	0.996436	8.59×10^{-5}
	$\sigma = 0.12$	0.962118	2.95×10^{-4}	0.946420	1.42×10^{-4}	0.990911	1.96×10^{-4}
	$\sigma = 0.18$	0.919507	3.93×10^{-4}	0.917845	3.19×10^{-4}	0.981927	1.73×10^{-4}
	$\sigma = 0.24$	0.868754	5.50×10^{-4}	0.887692	6.15×10^{-4}	0.968980	6.04×10^{-4}
	$\sigma = 0.30$	0.812142	9.42×10^{-4}	0.856005	1.21×10^{-3}	0.952211	5.21×10^{-4}
Finite projection	$N_{\text{shots}} = 10$	0.826176	4.53×10^{-4}	0.884200	1.22×10^{-3}	0.957392	4.55×10^{-4}
	$N_{\text{shots}} = 50$	0.957388	1.09×10^{-4}	0.949969	4.15×10^{-4}	0.989910	2.38×10^{-4}
	$N_{\text{shots}} = 100$	0.978063	1.37×10^{-4}	0.963715	1.68×10^{-4}	0.993988	9.32×10^{-5}
	$N_{\text{shots}} = 500$	0.995515	2.47×10^{-5}	0.978762	9.95×10^{-5}	0.997641	8.06×10^{-5}
	$N_{\text{shots}} = 1000$	0.997756	6.87×10^{-6}	0.981201	5.73×10^{-5}	0.998182	7.76×10^{-5}
	$N_{\text{shots}} = 5000$	0.999552	1.48×10^{-6}	0.983321	2.07×10^{-5}	0.998602	1.23×10^{-4}
Pepper readout	$p = 0.00$	0.999774	1.19×10^{-6}	0.983614	1.97×10^{-5}	0.998586	1.16×10^{-4}
	$p = 0.05$	0.967251	9.98×10^{-4}	0.939938	9.76×10^{-4}	0.997095	1.19×10^{-4}
	$p = 0.10$	0.934286	1.05×10^{-3}	0.899518	1.90×10^{-3}	0.995220	1.98×10^{-4}
	$p = 0.15$	0.901214	1.03×10^{-3}	0.857236	1.62×10^{-3}	0.992761	3.78×10^{-4}
	$p = 0.20$	0.869440	1.13×10^{-3}	0.820377	4.70×10^{-4}	0.989648	4.81×10^{-4}
	$p = 0.25$	0.835370	7.23×10^{-4}	0.780021	6.33×10^{-4}	0.984946	5.42×10^{-4}

2. Extended Scalability Results

This subsection reports the full numerical results for the trapped-ion-inspired scalability benchmark discussed in Section 3 C 3. For each qubit number $n \in \{2, 3, 4\}$, the mean fidelity and total inference time over the 1000-state test set were averaged across five independent runs.

To summarise the relative fidelity performance of the NN, the percentage fidelity improvement of the NN relative to a comparison method $X \in \{\text{Stokes}, \text{MLE}\}$ was defined as

$$I_F^{(\text{NN}, X)} = 100 \frac{\bar{F}_{\text{NN}} - \bar{F}_X}{\bar{F}_X}, \quad (\text{E5})$$

where $\bar{F}_{\text{NN}}$ and $\bar{F}_X$ are the corresponding mean fidelities. For inference time, the NN advantage

TABLE VII: Full numerical results for the trapped-ion-inspired scalability benchmark. Reported uncertainties are standard deviations over five runs.

n	Stokes		MLE		NN	
	$\bar{F}$	$\bar{T}$ (s)	$\bar{F}$	$\bar{T}$ (s)	$\bar{F}$	$\bar{T}$ (s)
2	0.95798 ± 0.00032	0.1834 ± 0.0018	0.86074 ± 0.00118	109.77 ± 1.77	0.99071 ± 0.00021	0.2282 ± 0.0147
3	0.90003 ± 0.00054	0.3686 ± 0.0018	0.78539 ± 0.00037	429.30 ± 3.04	0.98799 ± 0.00037	0.2631 ± 0.0674
4	0.79213 ± 0.00066	1.3333 ± 0.0063	0.71229 ± 0.00087	2287.20 ± 21.58	0.94916 ± 0.01813	0.2602 ± 0.0375

was expressed as a speedup factor rather than a percentage reduction,

$$S_T^{(\text{NN},X)} = \frac{\bar{T}_X}{\bar{T}_{\text{NN}}}, \quad (\text{E6})$$

where $\bar{T}_{\text{NN}}$ and $\bar{T}_X$ are the mean total inference times. Values $S_T > 1$ therefore indicate that the NN is faster than the comparison method, while $S_T < 1$ indicates that it is slower.

The uncertainty on the fidelity improvement was estimated by standard first-order propagation,

$$\sigma_{I_F} = 100 \sqrt{\left(\frac{\sigma_{F,\text{NN}}}{\bar{F}_X}\right)^2 + \left(\frac{\bar{F}_{\text{NN}}\sigma_{F,X}}{\bar{F}_X^2}\right)^2}, \quad (\text{E7})$$

where $\sigma_{F,\text{NN}}$ and $\sigma_{F,X}$ are the corresponding standard deviations of the mean fidelities. Similarly, for the speedup factor,

$$\sigma_{S_T} = S_T \sqrt{\left(\frac{\sigma_{T,X}}{\bar{T}_X}\right)^2 + \left(\frac{\sigma_{T,\text{NN}}}{\bar{T}_{\text{NN}}}\right)^2}, \quad (\text{E8})$$

where $\sigma_{T,\text{NN}}$ and $\sigma_{T,X}$ are the standard deviations of the mean total inference times.

The mean values reported across $n = 2\text{--}4$ in Table VIII were obtained by averaging the pointwise quantities over the three tested system sizes,

$$\bar{Q} = \frac{1}{N} \sum_{i=1}^N Q_i, \quad N = 3, \quad (\text{E9})$$

with uncertainty

$$\sigma_{\bar{Q}} = \frac{1}{N} \sqrt{\sum_{i=1}^N \sigma_{Q_i}^2}, \quad (\text{E10})$$

assuming independence between the propagated uncertainties at each qubit number.

The raw results confirm the trends discussed in the main text. The NN maintains the highest mean fidelity at all system sizes, while MLE exhibits the steepest degradation. Inference-time differences are even more pronounced: although the NN is slightly slower than Stokes at $n = 2$, it becomes faster by $n = 3$ and substantially faster by $n = 4$, while remaining several orders of magnitude faster than MLE throughout.

TABLE VIII: Relative performance of the NN against Stokes reconstruction and MLE in the trapped-ion-inspired scalability benchmark. Fidelity values are reported as percentage improvements in mean fidelity, while time values are reported as inference speedup factors.

Qubits	Fidelity improvement (%)		Inference speedup factor ($\times$)	
	NN vs Stokes	NN vs MLE	NN vs Stokes	NN vs MLE
2	3.42 ± 0.04	15.10 ± 0.16	0.80 ± 0.05	481 ± 32
3	9.77 ± 0.08	25.80 ± 0.08	1.40 ± 0.36	1632 ± 418
4	19.82 ± 2.29	33.25 ± 2.55	5.12 ± 0.74	8266 ± 1269
Mean across $n = 2-4$	11.00 ± 0.76	24.72 ± 0.85	2.44 ± 0.27	3635 ± 445